The bilinear Q4 element, derived from the lecture

MATH 6660 — companion note

From φ, ξ and $B(u,v)$ to $N_1,\dots,N_4$ and $\boldsymbol{k}^e$

0. First: a notation collision to get out of the way

The lecture uses ξ_i for the **nodal unknowns** (Lecture 13: $u_h = \sum_i \xi_i \varphi_i$).
Every Q4 textbook uses ξ, η for the **local element coordinates**.

These are unrelated objects and this note needs both, so from here on:

	symbol used here	what it is
nodal unknown	ξ_i	a *number* the solver computes
local coordinate	$(r,s) \in [-1,1]^2$	a *position* inside one element

If you are reading a textbook that writes (ξ,η) for the local coordinates, read my (r,s) as its (ξ,η).

The rest of the dictionary, which is the actual content of this note:

the lecturer writes	you write	they are
φ_i, basis function of V_h	N, shape function	**the same functions**, global vs. local view
ξ_i, coefficient in $u_h = \sum \xi_i \varphi_i$	u_i, nodal unknown	the same numbers
$B(u,v) = \iint \nabla u \cdot \nabla v$	virtual work	the same integral
$A_{ij} = B(\varphi_i, \varphi_j)$	$\boldsymbol{K}$, after assembly	the same matrix
$b_j = L(\varphi_j)$	$\boldsymbol{F}$, consistent nodal loads	the same vector
$V_h \subset V$	"the element is conforming"	the same requirement

Table 1: The dictionary this note is about: the same objects, named twice.

1. What the lecture actually demands

From Lecture 13, for 2D Poisson $-\Delta u = f$ on Ω, $u = 0$ on $\partial\Omega$:

$$B(u,v) = \iint_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega, \qquad L(v) = \iint_\Omega f v \, \mathrm{d}\Omega, \qquad V = H_0^1(\Omega), \tag{1}$$

and the method is: pick a finite-dimensional $V_h \subset V$, pick a basis $\{\varphi_i\}_{i=1}^M$ of it, write

$$u_h = \sum_{i=1}^{M} \xi_i \varphi_i, \tag{2}$$

and solve $\boldsymbol{A}\boldsymbol{\xi} = \boldsymbol{b}$ with $A_{ij} = B(\varphi_i, \varphi_j)$, $b_j = L(\varphi_j)$.

(2) **is already the engineering interpolation formula.** Compare it with what you write inside an element:

$$u_h = \sum_{i=1}^{M} \xi_i \varphi_i \qquad \text{vs.} \qquad u = \sum_{a=1}^{4} N_a u_a$$

Same shape. The *only* difference is the range of the sum: the lecture runs over all M **global** unknowns in the mesh, your code over the 4 **local** nodes of one element.

They agree because each φ has small support. At a point x inside element e, every term of the global sum dies except the four whose nodes belong to e, and those four φ's restrict to that element's four N's:

$$\sum_{i=1}^{M} \xi_i \varphi_i(x) = \sum_{a=1}^{4} u_a N_a(r,s) \quad \text{for } x \in K_e, \qquad \text{where } u_a = \xi_{\,\mathrm{conn}(e,a)}.$$

So (2) is your interpolation formula *before* it has been restricted to an element, and the connectivity table is what performs the restriction.

That is the *whole* specification. Everything below is forced by it. Three requirements, and each one makes a decision for us:

the lecture requires	because	so the element must
$V_h \subset H_0^1(\Omega)$	conforming (Lax–Milgram, Céa)	be C^0: no gaps between elements
$\iint \lVert\nabla v\rVert^2 < \infty$	$B(v,v)$ must be finite	have no jump in v (a jump $\Rightarrow \delta$ in ∇v)
$\{\varphi_i\}$ a *basis*	$\boldsymbol{A}$ must be invertible	have exactly one φ per unknown

Table 2: What the lecture's three requirements force the element to be.

Note what is *not* required: ∇v may jump across element edges. Only v itself must be continuous. That is the 2D version of Lecture 13's one-derivative requirement, and it is what makes a cheap element possible.

2. Choosing the polynomial: why *bi*linear

Take Ω meshed by rectangles. On one element we need a polynomial space. Count conditions:

- A quadrilateral has 4 corners. Put one unknown at each (the nodal value). So the element space must have **exactly 4 free coefficients** — one per unknown, or $\{\varphi_i\}$ is not a basis.
- **Why not the linear space** $\{1, x, y\}$? It has 3 coefficients, not 4. A plane through 4 prescribed corner values does not exist in general.
- **Why not the full quadratic** $\{1, x, y, x^2, xy, y^2\}$? That is 6, so 4 corner values would not determine it.
- So take $\{1,\ x,\ y,\ xy\}$ — **bilinear**, 4 coefficients:

$$v\big|_K = \alpha + \beta x + \gamma y + \delta\, xy. \tag{3}$$

Check it is C^0 — this is the step that earns the name. On the edge $y = \text{const}$, (3) reduces to $(\alpha + \gamma y) + (\beta + \delta y)x$, which is **linear in x alone**. A line is fixed by its two endpoints, and the

two elements sharing that edge share those two corner nodes — so they agree all along it. No gap. Hence $V_h \subset H_0^1$, and the element is conforming.

This is the xy term earning its keep. It is there to make the count 4, and it does *not* spoil continuity because it degenerates to a linear function on every edge. The lecture's triangle $\{1, x, y\}$ has 3 coefficients for 3 corners and is the triangular analogue of exactly this argument.

3. From φ to N: where the reference square comes from

The lecture never mentions N. It gives you φ. So start there and watch N appear.

Step 1 — what the lecture hands you

The mesh comes first. Elements exist from the very beginning: you cannot speak of "nodes" until Ω has been subdivided, and you cannot build V_h without that subdivision. The order is

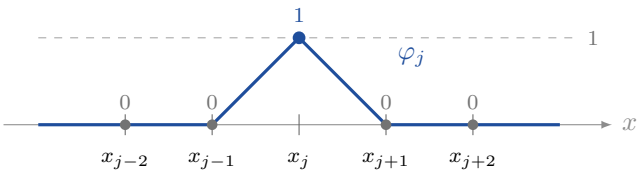
$$\text{mesh} \implies \text{nodes} \implies V_h = \{\text{piecewise polynomials on that mesh}\} \implies \text{basis } \{\varphi_i\}.$$

What has *not* appeared yet is the **reference** square and the local coordinates (r, s). Everything in this step is written in the real x, y of the real mesh.

The lecture's basis is the **nodal** one: one φ per global node, chosen so that

$$\varphi_j(x_k) = \delta_{jk} \qquad \text{for all } \textit{global} \text{ nodes } j, k = 1, \ldots, M. \tag{4}$$

In 1D, this is the picture the lecturer draws (Lectures 3–5, the "hat"):

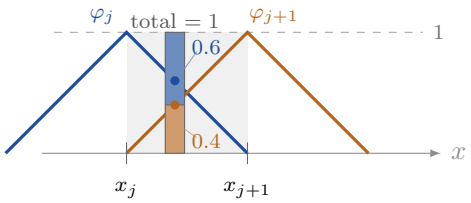


φ_j is 1 at its own node, 0 at every other node

Between the nodes, the surrounding nodes share the value. The nodal condition (4) pins φ_j down only *at* the nodes. At a point strictly inside an element more than one basis function is non-zero, and their values sum to one. On the element $[x_j, x_{j+1}]$ of width h exactly two hats survive,

$$\varphi_j(x) = \frac{x_{j+1} - x}{h}, \qquad \varphi_{j+1}(x) = \frac{x - x_j}{h}, \qquad \text{so} \qquad \varphi_j(x) + \varphi_{j+1}(x) = 1, \tag{5}$$

the **partition of unity**.

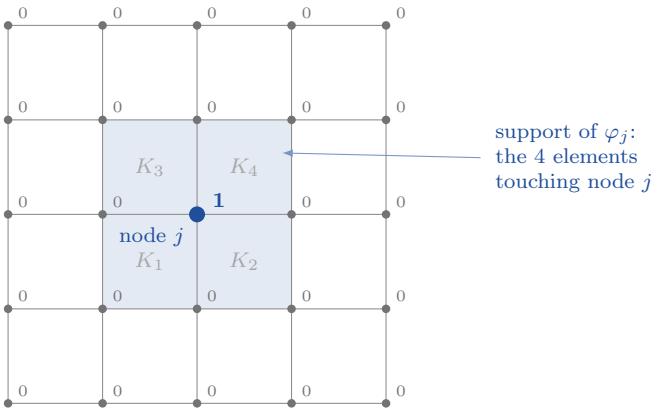


the two contributions stack to exactly 1 at every interior point

So $u_h(x) = \xi_j\varphi_j(x) + \xi_{j+1}\varphi_{j+1}(x)$ is a weighted average of the two nodal values whose weights sum to one: between nodes u_h is the straight line joining them, and it can never leave the range of its own nodal values. The Q4 element behaves identically with four contributions instead of two — that is (36).

The identity holds because the constant function 1 has nodal value 1 at every node and lies in V_h; since the nodal interpolant is unique, $\sum_j \varphi_j \equiv 1$. The sum must run over *all* nodes, boundary ones included — after the constrained boundary functions are dropped, the identity survives only away from $\partial\Omega$.

In 2D it is the same statement, now on a quadrilateral mesh. Below, the value of φ_j is written at every node; its support (shaded) is the four elements touching node j:



Both sketches say (4) and nothing else. Note what is *already* present in them: elements $(K_1, \ldots, K_4$ and all the others), nodes, and φ_j as a function of the real coordinates. Note what is absent: any mention of r, s, or a reference square. Those arrive in Step 3, and only as a computational convenience.

Step 2 — restrict to one element

Fix one element K. Two things happen:

- By §2, $\varphi_j\big|_K$ is a **bilinear polynomial** $\alpha + \beta x + \gamma y + \delta xy$.
- Of the M global basis functions, **only four are not identically zero on** K — those belonging to K's own four corners. Every other node lies outside K, and its φ has already fallen to zero there.

Write that out. Let the four corners of K be the global nodes j_1, j_2, j_3, j_4. Restricting the global sum (2) to K and separating those four terms from the rest:

$$u_h\big|_K = \sum_{i=1}^{M} \xi_i\,\varphi_i\big|_K = \underbrace{\xi_{j_1}\varphi_{j_1}\big|_K + \xi_{j_2}\varphi_{j_2}\big|_K + \xi_{j_3}\varphi_{j_3}\big|_K + \xi_{j_4}\varphi_{j_4}\big|_K}_{\text{the four corners of } K} + \sum_{\substack{i=1 \\ i\neq j_1,j_2,j_3,j_4}}^{M} \xi_i \underbrace{\varphi_i\big|_K}_{\equiv 0}. \tag{6}$$

Every term in the second sum vanishes identically on K: node i is not a corner of K, so K lies outside the support of φ_i. What is left is four terms,

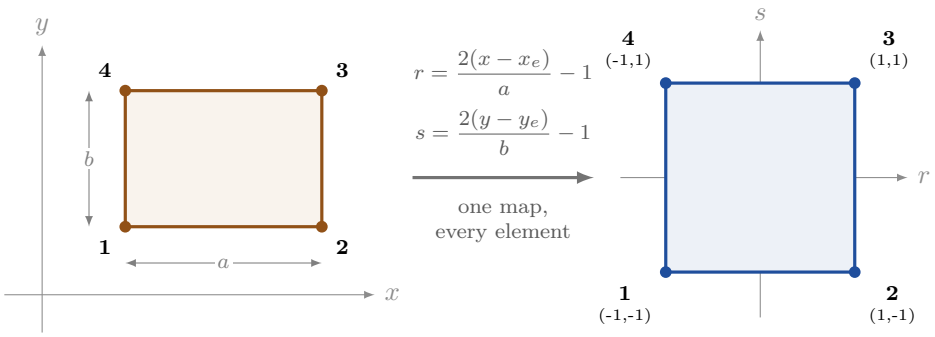
$$\boxed{u_h\big|_K = \sum_{a=1}^{4} \xi_{j_a}\,\varphi_{j_a}\big|_K} \tag{7}$$

so we need only four polynomials per element, not M.

Concretely, in the mesh drawn above there are $M = 25$ nodes, so (2) carries 25 terms. On the element K_1, 21 of them are identically zero and 4 survive. That ratio is what makes the method cheap — and it is why $\boldsymbol{A}$ is sparse.

Step 3 — map K to a fixed square

Each element has a different size and position, so $\varphi_j\big|_K$ is a *different* polynomial in x, y on every element. Rather than redo that algebra for each one, map every element onto a single **reference square** and do the work once:



element K (physical)
corner 1 at (x_e, y_e)

reference square
$(r, s) \in [-1, 1]^2$

Corner 1 goes to $(-1, -1)$, corner 2 to $(1, -1)$, corner 3 to $(1, 1)$, corner 4 to $(-1, 1)$. The numbering is *local* — $1, 2, 3, 4$ counter-clockwise from the lower-left corner *of that element*, which is a different global node in each element.

Why bother with (r, s) at all?

Because a real mesh contains elements of many sizes, positions and shapes, and the *same* local shape function looks different on each one. Figure 1 shows four of them.

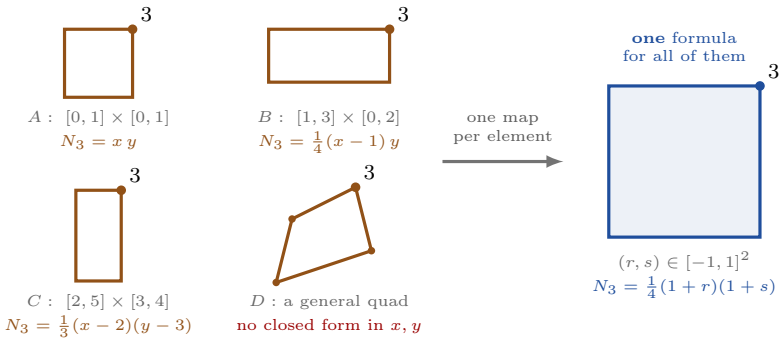


Figure 1: The same local shape function — the one that is 1 at corner 3 and 0 at the other three — on four different elements. In x, y it is a different polynomial every time, and on the distorted quad D there is no convenient closed form at all. Mapped to the reference square it is a single formula, identical for every element in the mesh.

The numbers. Fix one point of the reference square, say $(r, s) = (0.5, 0.5)$, and ask: *how much does node* 3 *contribute there?* The answer is $N_3(0.5, 0.5) = \frac{1}{4}(1.5)(1.5) = 0.5625$. That one reference point sits at a *different physical place* (x, y) in each element A, B, C, D, and in x, y it is evaluated by a *different formula* in each — yet every one returns the same share.

The other three make up the rest. Node 3 does not act alone: at the same point

$$\underbrace{N_3(0.5,0.5)=0.5625}_{\text{node 3}} \;+\; \underbrace{N_1+N_2+N_4=0.0625+0.1875+0.1875=0.4375}_{\text{the other three nodes}} \;=0.5625{+}0.4375=\;1, \tag{8}$$

which is the partition of unity (36) again, now read as a statement about *shares*: the four nodes divide the point between them, and the division is the same whether it is described in (r,s) or in x,y.

The irony in row D. For D we cannot write down equations for $N_1(x,y)$, $N_2(x,y)$, $N_3(x,y)$ or $N_4(x,y)$ *at all* — not one of the four has a closed form in physical coordinates. And suppose we could: each would be particular to that one element, so a mesh of 10,000 elements would carry

$$4 \text{ shape functions} \times 10{,}000 \text{ elements} \;=\; 40{,}000 \text{ formulas},$$

against exactly 4 in (r,s).

And it is exactly what Gauss quadrature needs. Quadrature rules are not derived per problem — they are *tabulated*, once, on the standard interval $[-1,1]$. The n-point Gauss–Legendre rule supplies abscissae r_q and weights w_q with

$$\int_{-1}^{1} g(r)\,\mathrm{d}r \;\approx\; \sum_{q=1}^{n} w_q\, g(r_q), \qquad \text{exact for polynomials of degree} \le 2n-1. \tag{9}$$

Because the reference element *is* $[-1,1]^2$, those tabulated numbers apply unchanged to every element in the mesh. Taking the product rule in two dimensions and inserting the Jacobian,

$$\iint_K f\,\mathrm{d}\Omega = \int_{-1}^{1}\!\int_{-1}^{1} f\big(x(r,s),y(r,s)\big)\,|J|\,\mathrm{d}r\,\mathrm{d}s \;\approx\; \sum_{q} w_q\, f\big(x(r_q,s_q),y(r_q,s_q)\big)\,\big|J(r_q,s_q)\big|. \tag{10}$$

For the Q4 element the 2×2 rule — four points at $(r,s)=(\pm\frac{1}{\sqrt{3}},\pm\frac{1}{\sqrt{3}})$, all four weights equal to 1 — is enough, and *on a rectangle it is exact*: evaluating (41) that way reproduces (43) to machine precision, because the integrand is degree 2 in each variable separately while the rule is exact through degree 3.

Where that degree 2 comes from — in the $\boldsymbol{B}$-matrix form you already use. Write the Q4 element stiffness as it appears in solid mechanics,

$$\boldsymbol{k}^e = \underbrace{\int_{\Omega^e} \boldsymbol{B}^T(x,y)\,\boldsymbol{D}\,\boldsymbol{B}(x,y)\,\mathrm{d}\Omega}_{\text{over the physical element}} = \underbrace{\int_{-1}^{1}\!\int_{-1}^{1} \boldsymbol{B}^T(r,s)\,\boldsymbol{D}\,\boldsymbol{B}(r,s)\,\big|J(r,s)\big|\,\mathrm{d}r\,\mathrm{d}s}_{\text{over the reference square}}, \tag{11}$$

with $\boldsymbol{d}=[u_1\,v_1\,u_2\,v_2\,u_3\,v_3\,u_4\,v_4]^T$, $\boldsymbol{\varepsilon}=\boldsymbol{B}\boldsymbol{d}$ and, for plane stress, $\boldsymbol{D}=\frac{E}{1-\nu^2}\begin{bmatrix}1&\nu&0\\ \nu&1&0\\ 0&0&\frac{1-\nu}{2}\end{bmatrix}$. The strain–displacement matrix is $\boldsymbol{B}=\partial\boldsymbol{N}$:

$$\boldsymbol{B}=\begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} \end{bmatrix}. \tag{12}$$

One caution on the notation in (11). Writing $\boldsymbol{B}(r,s)$ does *not* mean its entries are derivatives with respect to r and s. They are still $\partial N_a/\partial x$ and $\partial N_a/\partial y$ — derivatives with respect to the *physical*

coordinates, as (12) says. The argument (r, s) only records *where* they are evaluated. The two are connected by the chain rule (23): the code forms $\partial N_a/\partial r$ and $\partial N_a/\partial s$ first because those are the easy ones, then multiplies by J^{-1} to get the x, y derivatives that $\boldsymbol{B}$ actually holds.

The summation form — what the code actually loops over. The integral in (11) is never evaluated symbolically. Replacing it by the Gauss sum (10) gives the matrix form

$$\boxed{\boldsymbol{k}^e = \sum_{q=1}^{n_{gp}} w_q \, \boldsymbol{B}^T(r_q, s_q)\, \boldsymbol{D}\, \boldsymbol{B}(r_q, s_q)\, \big|J(r_q, s_q)\big|} \tag{13}$$

and, written entry by entry, exactly the nested loops a program runs:

$$k^e_{ij} = \sum_{q=1}^{n_{gp}} w_q \, \big|J_q\big| \sum_{k=1}^{3}\sum_{l=1}^{3} B_{ki}(r_q, s_q)\; D_{kl}\; B_{lj}(r_q, s_q), \qquad i, j = 1, \ldots, 8. \tag{14}$$

For the Q4 element $n_{gp} = 4$, with $(r_q, s_q) = (\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}})$ and $w_q = 1$.

In code that is four loops — over Gauss points, then over the 8×8 entries, with the 3×3 material contraction inside:

```
ke = zeros(8,8)
for q in 1..ngp:      # Gauss points
  r, s, w = gp[q]
  dNdr, dNds = shape_derivs(r, s)      # 4 each
  Jm = [[dNdr·x, dNdr·y], [dNds·x, dNds·y]]      # 2x2
  detJ = det(Jm)
  dNdx, dNdy = solve(Jm, [dNdr; dNds])      # chain rule
  B = assemble_B(dNdx, dNdy)      # 3x8, eq. 12
  ke += w * detJ * B.T @ D @ B
```

Note the order: the shape-function derivatives are formed in (r, s), the Jacobian converts them to x, y, and only then is $\boldsymbol{B}$ built. The element's geometry enters through `Jm` alone — `shape_derivs` is the same four formulas for every element in the mesh.

Now read the degrees straight off (11):

- **Every entry of $\boldsymbol{B}$ is a *first derivative* of a shape function.** N_a is degree 1 in r and 1 in s; differentiating removes one, so each $N_{a,x}$ or $N_{a,y}$ is degree ≤ 1 in r and ≤ 1 in s.
- **$\boldsymbol{D}$ contributes no r, s dependence** — for a homogeneous linear-elastic material it is a matrix of constants.
- **So $\boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}$ is $\boldsymbol{B}$ *times* $\boldsymbol{B}$.** Each entry is $\sum_{k,l} B_{ki} D_{kl} B_{lj}$ — a sum of *products of two* $\boldsymbol{B}$ entries, i.e. two first derivatives multiplied together. One derivative is degree ≤ 1 per variable; two multiplied give degree ≤ 2 per variable.

That is what "the integrand multiplies two derivatives" meant: it is the $\boldsymbol{B}^T(\cdots)\boldsymbol{B}$ sandwich. Checked symbolically on the unit square: every entry of $\boldsymbol{B}$ has degree ≤ 1 in r and in s, and every entry of $\boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}$ has degree exactly 2 in r and 2 in s — so the 2-point rule, exact through degree 3, covers it.

The Poisson problem of this note is the same structure with one unknown per node instead of two: $\boldsymbol{B}$ collapses to the 2×4 gradient matrix $\begin{bmatrix} N_{1,x} & N_{2,x} & N_{3,x} & N_{4,x} \\ N_{1,y} & N_{2,y} & N_{3,y} & N_{4,y} \end{bmatrix}$ and $\boldsymbol{D} = \boldsymbol{I}$, so $\boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}$ reduces entry-by-entry to $\nabla N_a \cdot \nabla N_b = N_{a,x}N_{b,x} + N_{a,y}N_{b,y}$ — the integrand in (41). Two derivatives multiplied, again.

On a distorted element it is no longer exact. Everything above assumed the integrand is a *polynomial*. Here is where that assumption comes from, and where it breaks.

Step 1 — the chain rule divides by $\det J$**.** The easy derivatives are $\partial N_a/\partial r$ and $\partial N_a/\partial s$. The ones $\boldsymbol{B}$ needs are $\partial N_a/\partial x$ and $\partial N_a/\partial y$, and they are obtained by *inverting* the Jacobian:

$$\underbrace{\begin{bmatrix} N_{1,x} & N_{2,x} & N_{3,x} & N_{4,x} \\ N_{1,y} & N_{2,y} & N_{3,y} & N_{4,y} \end{bmatrix}}_{\text{the } x,y \text{ derivatives, } 2\times 4} = \boldsymbol{J}^{-1} \underbrace{\begin{bmatrix} N_{1,r} & N_{2,r} & N_{3,r} & N_{4,r} \\ N_{1,s} & N_{2,s} & N_{3,s} & N_{4,s} \end{bmatrix}}_{\text{the easy ones, } 2\times 4}, \qquad \boldsymbol{J}^{-1} = \frac{1}{\det \boldsymbol{J}} \operatorname{adj} \boldsymbol{J}. \tag{15}$$

Those eight numbers are precisely what fills the element strain–displacement matrix: row 1 of (15) supplies the top row of $\boldsymbol{B}^e$, row 2 supplies the middle row, and the two interleaved give the shear row,

$$\boldsymbol{B}^e = \begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} \end{bmatrix} \qquad \text{(this is (12))}. \tag{16}$$

Since every $N_{a,x}$ and $N_{a,y}$ came out of (15) carrying the factor $1/\det \boldsymbol{J}$ from $\boldsymbol{J}^{-1}$, **every entry of** $\boldsymbol{B}^e$ **carries it too**.

Step 2 — count the factors in the integrand. From (11) the integrand is $\boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\, |J|$:

$$\underbrace{\frac{1}{\det \boldsymbol{J}}}_{\text{from the first } \boldsymbol{B}} \times \underbrace{\frac{1}{\det \boldsymbol{J}}}_{\text{from the second } \boldsymbol{B}} \times \underbrace{\det \boldsymbol{J}}_{\text{from } \mathrm{d}\Omega} = \frac{1}{\det \boldsymbol{J}}.$$

One factor cancels, **one survives in the denominator**.

Step 3 — so it depends entirely on whether $\det \boldsymbol{J}$ **is constant.** First, what $\boldsymbol{J}$ actually is. It is the 2×2 matrix of derivatives of the map, and because the map is $x = \sum_a N_a x_a$, $y = \sum_a N_a y_a$, it is built straight from the nodal coordinates:

$$\boldsymbol{J} = \begin{bmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial y}{\partial r} \\ \dfrac{\partial x}{\partial s} & \dfrac{\partial y}{\partial s} \end{bmatrix} = \begin{bmatrix} \sum_a N_{a,r}\, x_a & \sum_a N_{a,r}\, y_a \\ \sum_a N_{a,s}\, x_a & \sum_a N_{a,s}\, y_a \end{bmatrix}, \qquad \det \boldsymbol{J} = \frac{\partial x}{\partial r}\frac{\partial y}{\partial s} - \frac{\partial y}{\partial r}\frac{\partial x}{\partial s}. \tag{17}$$

(That second form is the `Jm` line of the loop above.) Now put the two elements through it.

- **Rectangle** $a \times b$. The map is simply $x = x_e + \frac{a}{2}(r+1)$, $y = y_e + \frac{b}{2}(s+1)$ — x depends on r alone and y on s alone, so the off-diagonal derivatives vanish:

$$\boldsymbol{J} = \begin{bmatrix} \frac{a}{2} & 0 \\ 0 & \frac{b}{2} \end{bmatrix}, \qquad \det \boldsymbol{J} = \frac{a}{2} \cdot \frac{b}{2} - 0 \cdot 0 = \frac{ab}{4}.$$

  No r**, no** s **anywhere in it** — $\det \boldsymbol{J}$ is a *number*, fixed by the element's two side lengths. Dividing a polynomial by a number leaves a polynomial, so Gauss is exact.

Why $ab/4$ and not ab? $\det \boldsymbol{J}$ is *not* the area — it is the **area scaling factor**, the ratio of physical area to reference area:

$$\mathrm{d}\Omega = \mathrm{d}x\,\mathrm{d}y = \det \boldsymbol{J}\ \mathrm{d}r\,\mathrm{d}s.$$

The reference square $[-1,1]^2$ has area $2\times 2 = \mathbf{4}$, not 1. So the area is recovered by *integrating*, not by reading:

$$\text{area} = \int_{-1}^{1}\!\int_{-1}^{1} \det \boldsymbol{J}\,\mathrm{d}r\,\mathrm{d}s = \frac{ab}{4}\times 4 = ab. \ \checkmark$$

A second way to see that $\det \boldsymbol{J}$ cannot itself be an area: on element D it *varies* from 3 to 4 across the element, and one element has only one area. Its integral is the area, though — $\iint \det \boldsymbol{J} = 14$, exactly D's area by the shoelace formula.

- **Distorted quad D.** Here the map (25) is $x = 2 + \frac{3}{2}r + \frac{1}{2}s$, $y = \frac{9}{4} - \frac{1}{4}r + \frac{9}{4}s - \frac{1}{4}rs$ — note the rs term, which makes y depend on r as well. So the off-diagonal entries do *not* vanish:

$$\boldsymbol{J} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{4} - \frac{s}{4} \\ \frac{1}{2} & \frac{9}{4} - \frac{r}{4} \end{bmatrix}, \qquad \det \boldsymbol{J} = \tfrac{3}{2}\Big(\tfrac{9}{4} - \tfrac{r}{4}\Big) - \Big(-\tfrac{1}{4} - \tfrac{s}{4}\Big)\Big(\tfrac{1}{2}\Big) = \tfrac{7}{2} - \tfrac{3r}{8} + \tfrac{s}{8}.$$

Now r and s survive — $\det \boldsymbol{J}$ is a *function*, taking a different value at every point of the element. It stays in the denominator, and the integrand becomes a **ratio of polynomials**:

$$\frac{\partial N_1}{\partial x} = \frac{r + 4s - 5}{28 - 3r + s} \qquad (\text{denominator} = 8\det \boldsymbol{J}),$$

$$\underbrace{\Big[\big(\tfrac{\partial N_1}{\partial x}\big)^2 + \big(\tfrac{\partial N_1}{\partial y}\big)^2\Big]\det \boldsymbol{J}}_{\text{the } k_{11}^e \text{ integrand}} = \frac{(r+4s-5)^2 + (3r - s - 2)^2}{8\,(28 - 3r + s)}. \tag{18}$$

Four Gauss points, four different $\det \boldsymbol{J}$ — what does that mean? The 2×2 rule puts its points at $(r, s) = (\pm g, \pm g)$ with $g = 1/\sqrt{3} = 0.577350$, all weights 1. Evaluating $\det \boldsymbol{J} = \frac{7}{2} - \frac{3r}{8} + \frac{s}{8}$ at each:

q	(r_q, s_q)	maps to (x, y)	$\det \boldsymbol{J}$	value
1	$(-g, -g)$	(0.8453, 1.0120)	$\frac{7}{2} + \frac{\sqrt{3}}{12}$	3.644338
2	$(+g, -g)$	(2.5774, 0.8900)	$\frac{7}{2} - \frac{\sqrt{3}}{6}$	3.211325
3	$(+g, +g)$	(3.1547, 3.3214)	$\frac{7}{2} - \frac{\sqrt{3}}{12}$	3.355662
4	$(-g, +g)$	(1.4226, 3.7767)	$\frac{7}{2} + \frac{\sqrt{3}}{6}$	3.788675

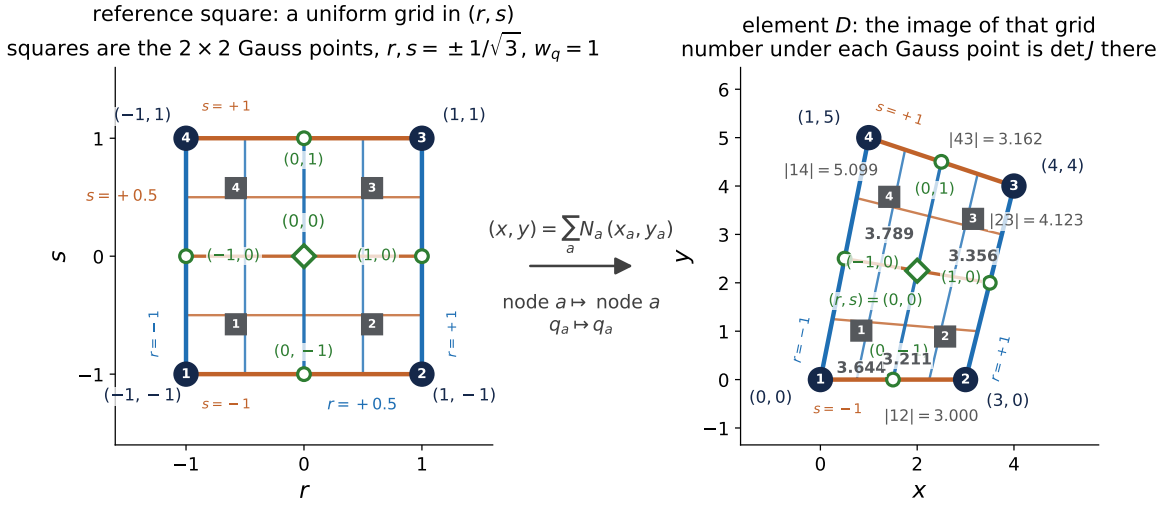


Figure 2: Left: the reference square. **Right:** element D, carrying the *same* grid. Blue lines are r = const, orange lines are s = const, and each edge is labelled with the grid line it *is*; the two middle members are the bimedians. Every marker carries its index *inside* it, so node a and Gauss point q_a mean the same thing in both panels. Green: (r, s)-landmarks — the centre and the four parent edge midpoints — and their images. Grey: edge lengths $|ab|$, and $\det \boldsymbol{J}$ under each Gauss point.

The four parent corners, put through the map. At a corner three of the four N_a vanish and the fourth is 1 — the nodal condition (27) — so the sum collapses onto a single nodal coordinate:

(r, s)	(N_1, N_2, N_3, N_4)	$x = \sum_a N_a x_a$	$y = \sum_a N_a y_a$	gives
$(-1, -1)$	$(1, 0, 0, 0)$	$1(0) + 0(3) + 0(4) + 0(1)$	$1(0) + 0(0) + 0(4) + 0(5)$	$(0, 0) =$ node 1
$(1, -1)$	$(0, 1, 0, 0)$	$0(0) + 1(3) + 0(4) + 0(1)$	$0(0) + 1(0) + 0(4) + 0(5)$	$(3, 0) =$ node 2
$(1, 1)$	$(0, 0, 1, 0)$	$0(0) + 0(3) + 1(4) + 0(1)$	$0(0) + 0(0) + 1(4) + 0(5)$	$(4, 4) =$ node 3
$(-1, 1)$	$(0, 0, 0, 1)$	$0(0) + 0(3) + 0(4) + 1(1)$	$0(0) + 0(0) + 0(4) + 1(5)$	$(1, 5) =$ node 4

Reading the same four off the collapsed form (25) costs no N's at all: $x = 2 + \frac{3r}{2} + \frac{s}{2}$ and $y = \frac{9}{4} - \frac{r}{4} + \frac{9s}{4} - \frac{rs}{4}$ give, at $(-1, -1)$, $2 - \frac{3}{2} - \frac{1}{2} = 0$ and $\frac{9}{4} + \frac{1}{4} - \frac{9}{4} - \frac{1}{4} = 0$, and so on round the element.

What that table is telling you.

1. The element passes exactly through its own nodes. This is the one place where the map is not an approximation of anything: corner a of the parent lands *on* node a, for any quadrilateral you like, however distorted. It is (27) doing its job in the geometry slot instead of the unknown slot.

2. Everywhere else all four are alive, and they are weights. They sum to 1 at every point of the square, (36), so the image is always a *weighted average* of the four corners — which is also why it can never escape the quad. At the centre all four equal $\frac{1}{4}$:

$$x(0,0) = \tfrac{0+3+4+1}{4} = 2, \qquad y(0,0) = \tfrac{0+0+4+5}{4} = \tfrac{9}{4},$$

the green diamond in the figure. The corner table is just the extreme case of this, with the weights $(1,0,0,0)$ instead of $(\frac{1}{4},\frac{1}{4},\frac{1}{4},\frac{1}{4})$.

3. On an edge, two of them die and the other two go linear. Set $s=-1$:

$$(N_1,N_2,N_3,N_4)\big|_{s=-1} = \big(\tfrac{1-r}{2},\ \tfrac{1+r}{2},\ 0,\ 0\big).$$

Nodes 3 and 4 drop out completely and the survivors are linear in r, so that edge maps to the *straight* segment from node 1 to node 2 and the parent midpoint $(0,-1)$ lands on the physical midpoint $(\frac{3}{2},0)$ — the green ring on the bottom edge. Two consequences worth keeping: *(i)* the edges of a Q4 are straight even though its interior map is bilinear — every N_a carries one factor $(1\pm r)$ and one $(1\pm s)$, so fixing $s=\pm1$ kills two of them outright and strips the s out of the other two, leaving a map linear in r alone; *(ii)* the image of edge $s=-1$ depends on nodes 1 and 2 **and nothing else**, so the element on the far side of that edge — whatever its own local numbering — builds the identical segment from the identical two nodes. That is the conformity of §2, visible here as two zeros in a row of the table.

4. One thing to be careful about. $(0,0)$ maps to the plain average of the four corners, $(2,\frac{9}{4})$. That is the *vertex* centroid, and D has two other equally reasonable "centres" — all three different:

$$\underbrace{\big(2,\tfrac{9}{4}\big)}_{\text{vertex centroid}=(0,0)}, \qquad \underbrace{\big(\tfrac{41}{21},\tfrac{16}{7}\big)}_{\text{area centroid}} = (1.9524, 2.2857), \qquad \underbrace{\big(\tfrac{15}{7},\tfrac{15}{7}\big)}_{\text{diagonals cross}} = (2.1429, 2.1429).$$

They coincide for a parallelogram and separate for everything else, so "the centre of the element" is ambiguous unless you say which one you mean. Everything in this note that gets evaluated at $(r,s)=(0,0)$ — the map, the crossing point of the two bimedians, the mean of the four corner Jacobians — returns the first.

"Conforming" — and the two other things that word is used for. "Continuous" and "conforming" are not two words for the same thing, and that is the whole source of the confusion. They answer different questions.

Two questions, four boxes.

the word	the question it answers	where you look
continuous / discontinuous	is there a *gap* at the shared edge?	at the mesh
conforming / non-conforming	is that gap *legal* for this equation?	at the PDE

Table 3: The two independent questions behind the four combinations of Figure 4. The second one is not a property of the element: change the PDE and the answer changes without the element moving.

Two PDEs, the same ten steps. Only one thing differs between the columns — how many times you integrate by parts. Everything under that row is forced by it.

	Poisson, order 2	**Euler–Bernoulli beam**, order 4
strong form	$-(k\,u')' = f$	$(EI\,w'')'' = q$
integrate by parts	*once*	*twice*
weak form	$\int k\,u'v' = \int fv$	$\int EI\,w''v'' = \int qv$
derivatives, trial / test	$1\,/\,1$	$2\,/\,2$
so the space is	$V = H^1$	$V = H^2$
must be continuous at the edge	u	w *and* w'
free to jump	u'	w''
continuity class	C^0	C^1
what the two elements share	the value	the value *and* the slope
cheapest conforming element	linear Lagrange, 2 DOF	Hermite cubic, 4 DOF

Table 4: The same derivation run on a second- and a fourth-order problem. How the derivatives split between trial and test function is what fixes V; V fixes the continuity class; and the continuity class fixes which quantities have to be nodal unknowns.

The pattern. A PDE of order $2m$ hands m derivatives to the trial function and m to the test function, so $V = H^m$, so the element must be C^{m-1}: everything up to the $(m-1)$st derivative is shared at the node, and the mth derivative is free to jump. That one sentence generates both columns.

One trap: C^1 does not mean "quadratic". Continuity class and polynomial degree are independent. A quadratic Lagrange element is still C^0 — the extra mid-side node raises the *degree*, but the element still shares only *values* with its neighbour, so the slope still jumps and a beam built from it is still non-conforming. C^1 comes from making the slope a *nodal unknown*, which is exactly what the Hermite cubic does: degree 3, and C^1 because w' sits at the node and both elements read the same one. Raising the degree never buys continuity; changing which quantities are shared does.

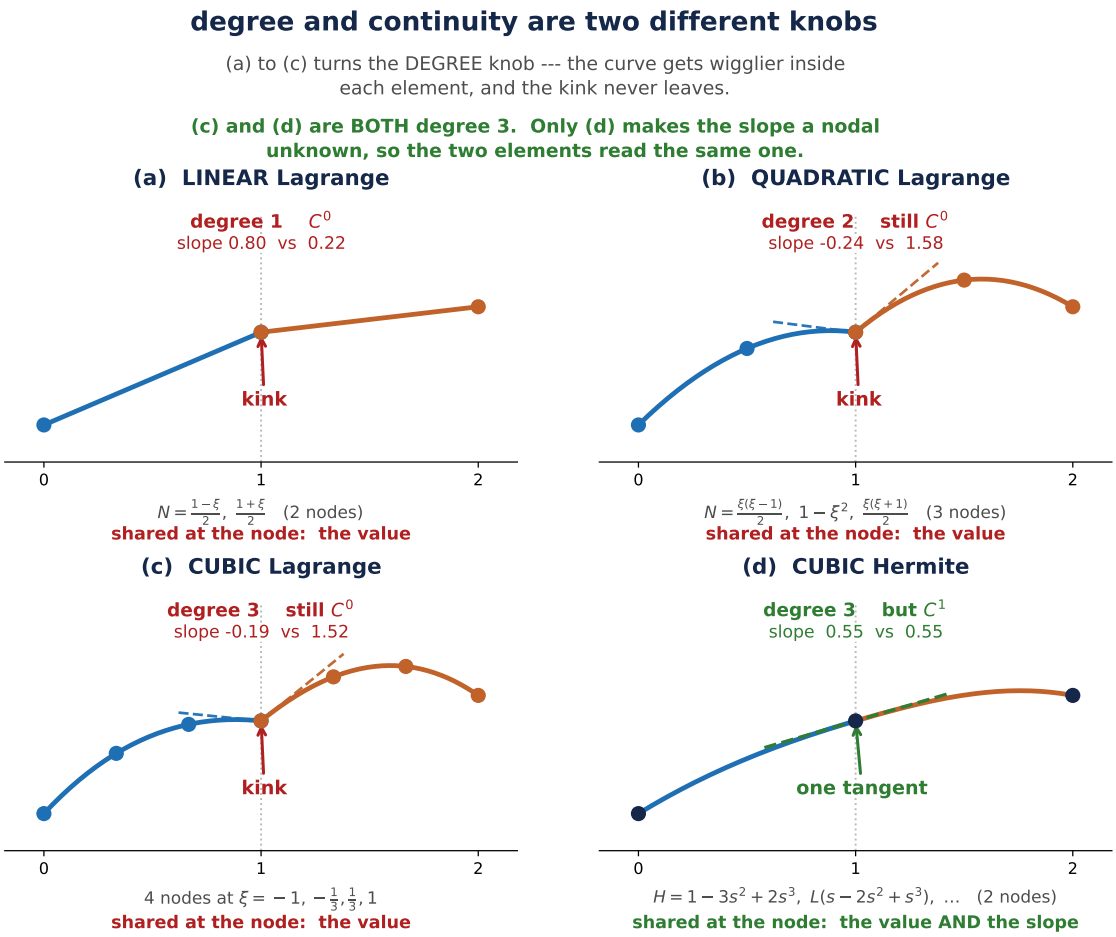


Figure 3: Degree and continuity are two separate knobs, shown on two elements joined at $x = 1$. **(a) → (b) → (c)** turns the *degree* knob — 1, 2, 3 — and the curve gets wigglier inside each element while the kink at the joint never leaves: all three are C^0, because all three share only the *value* with the neighbour. The dashed tangents and the printed slopes make the mismatch explicit (0.80 vs 0.22, then −0.24 vs 1.58, then −0.19 vs 1.52). **(c) and (d) are both degree** 3 — the comparison worth staring at. (d) is C^1 not because it is a higher polynomial, but because w' was made a *nodal unknown*, so the two elements read one tangent instead of two. **Bottom strip:** the same four elements judged by the two PDEs of Table 4. Poisson asks for C^0 and every column passes; the beam asks for C^1 and only (d) does. Turning the degree knob moves nothing in either row.

Q. What if I use C^0 linear Lagrange on the Euler–Bernoulli beam — is that conforming or non-conforming? look at the pde only. no need to look at the mesh

A. Non-conforming — and not in a useful way. The element has *zero* bending stiffness.

Follow the arithmetic rather than the definition. A linear w has $w'' = 0$ inside the element, so the weak-form integrand $EI\, w''v''$ is identically zero and

$$k^e = \int_0^L EI\, w''\, v''\, \mathrm{d}x = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \text{rank} = 0.$$

Not nearly singular — *exactly* the zero matrix. Every nodal displacement is a zero-energy mode, the assembled K is entirely singular, and the solver has nothing to invert.

The formal reason is that same fact seen from the other side. Piecewise-linear C^0 functions are not in H^2: the slope jumps at every node, so w'' carries a Dirac delta there and $\int (w'')^2$ is not finite. The energy is 0 inside the elements and undefined at the nodes, so $V_h \not\subset V$.

So the same element on the same mesh — C^0 linear Lagrange — is *conforming* for Poisson and *non-conforming* for the beam. Nothing about the element changed; the equation did. That is the whole distinction.

What the weak form does and does not settle. Counting the derivatives in the weak form gives you the *requirement*: order $2m$ needs C^{m-1}, and you can read that off without ever seeing a mesh. It does not give you the *verdict*. Whether a particular element meets that bar is a question about the element — which quantities does it actually share with its neighbour? Two steps, two different objects, which is precisely the two rows of Table 3. Crouzeix–Raviart is the reminder: a second-order problem asks only for C^0, and that element still fails to deliver it, being continuous at edge midpoints alone.

Q. Is C^1 possible at degree 1 or 2? And could I use it on a beam, or is it only H^1?

A. As a *nodal element*, no — the minimum is cubic. As a *spline*, yes — and it does carry a beam. The phrase "C^1 quadratic" means two different things, and they have opposite answers.

As a nodal element: count coefficients against conditions. C^1 means matching the value *and* the slope at *both* ends: four demands.

degree	free coefficients	four conditions?	what you get
1, linear	2	×	the whole mesh collapses to one line
2, quadratic	3	×	the nodal values are not independent
3, cubic	4	✓	the Hermite beam element

Table 5: Why the Hermite element is *cubic*. It is not a design choice — four conditions need four coefficients.

For the quadratic the system has no solution: impose the first three conditions and the fourth forces

$$w_2 = w_1 + \tfrac{L}{2}\big(\theta_1 + \theta_2\big),$$

so you cannot hand such an element an arbitrary $(w_1, \theta_1, w_2, \theta_2)$. Degree 1 fails worse, and for a reason worth picturing: if a piecewise-linear function has a continuous slope at *every* node, the slope is the same everywhere and the whole mesh is one straight line — two degrees of freedom in total, however many elements you use. The rule behind the table is that C^k needs degree $\geq 2k+1$: $C^0 \to 1$, $C^1 \to 3$, $C^2 \to 5$.

As a spline: a quadratic *is* C^1, and it works. Quadratic B-splines with simple knots are C^1 across every knot. Checked on four elements: the jump in w' at an interior knot is 1.6×10^{-6} (zero),

while the jump in w'' is 48 — so exactly C^1, not C^2. Assembling the beam stiffness $\int EI\, w''v''\,\mathrm{d}x$ over that space gives eigenvalues

$$0, \quad 0, \quad 50.49, \quad 363.2, \quad 973.5, \quad 1172.8,$$

rank 4 of 6, nullity 2 — the two rigid-body modes, exactly as for the Hermite cubic, and emphatically *not* the zero matrix that C^0 linear Lagrange produced above.

How does that dodge the counting argument? It does not have four independent degrees of freedom per element. On four elements the entire space carries 6 unknowns, not 16: DOF $= N + p$. The basis functions straddle several elements and their coefficients are shared control points rather than nodal values, so continuity is built into the *basis* instead of being enforced element by element. That is isogeometric analysis, and it is why IGA can do Kirchhoff–Love shells without Hermite degrees of freedom. The price is that the unknowns are control points: you cannot read a nodal deflection off the solution vector, and boundary conditions become awkward.

So, the H^1 question. Both rows below are degree 2:

degree-2 space	lives in	usable on a beam?
C^0, 3-node Lagrange	H^1, *not* H^2	$\times$ w'' carries deltas; the energy is undefined
C^1, quadratic B-spline	H^2	✓ w'' is piecewise constant, hence in L^2

Table 6: H^1 versus H^2 is decided by the continuity class, not by the polynomial degree — both of these are quadratics.

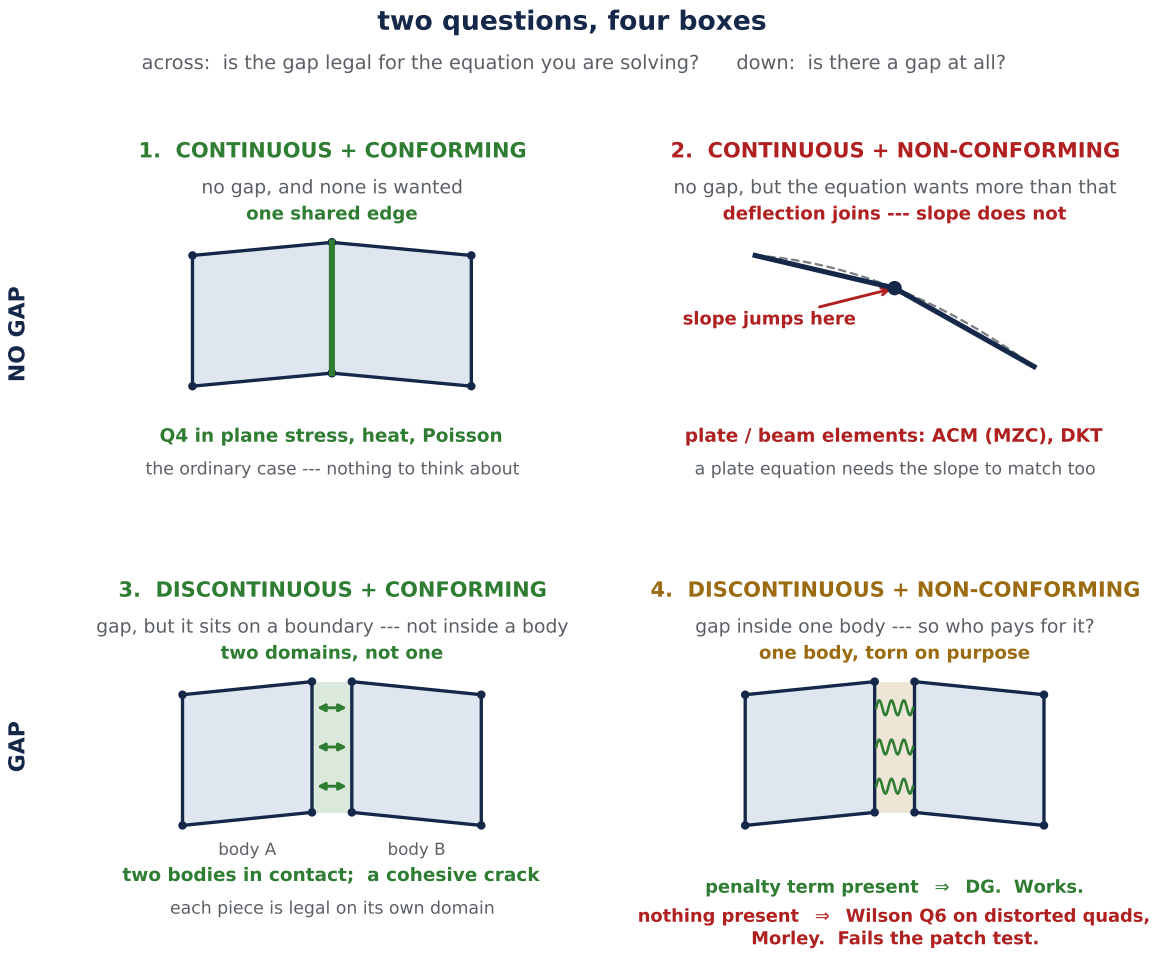


Figure 4: The four combinations, all drawn on the same two elements. **1:** the ordinary case. **2:** no gap at all, yet still non-conforming — a plate equation asks for the *slope* to match too, and it does not. **3:** a real gap that is still legal, because it lies on a boundary between two bodies rather than inside one. **4:** a gap inside one body, where everything depends on whether the weak form has a term that pays for it — a penalty (DG, which works) or nothing at all (Wilson Q6 on distorted quads, Morley, which fails the patch test).

The definition is one line: $V_h \subset V$. The space you actually compute in has to be a genuine *subspace* of the space the weak form lives in. Note what that is not: it is not a property of an element on its own. A single Q4 is neither conforming nor non-conforming. It is a statement about how two elements *join*.

What it costs depends on the order of the PDE, because that is what fixes V:

the problem	V	conforming means	so the element is
bar, heat, plane stress, Poisson	H^1	C^0: v continuous, ∇v *may* jump	Q4, T3 — cheap
beam, plate, biharmonic	H^2	C^1: v *and* ∇v continuous, v'' *may* jump	Hermite in 1D; Argyris, 21 DOF, in 2D

Your cantilever is the second row. CG(3) Hermite carries the value *and* the slope as nodal unknowns, and both are shared by the two elements meeting at a node — so u_h is C^1 and $V_h \subset H^2$. The price of the same thing in 2D is Lecture 13's Argyris element: 21 degrees of freedom on one triangle, second derivatives carried as nodal unknowns.

Why the lecture cares. If $V_h \not\subset V$ then $B(v_h, v_h)$ is not merely wrong, it is *undefined*: a jump in v puts a δ in ∇v and $\iint \|\nabla v\|^2 = \infty$. Lax–Milgram, Galerkin orthogonality and Céa are all stated on $V_h \subset V$. Lose it and every error estimate in Lecture 13 goes with it — which is why the lecture spends 21 degrees of freedom to keep it.

And that is what the two zeros were. On $s = -1$, $(N_1, N_2, N_3, N_4) = \left(\frac{1-r}{2}, \frac{1+r}{2}, 0, 0\right)$. The zeros say the trace on that edge cannot possibly depend on nodes 3 and 4 — the two the neighbour does *not* share. So both elements build the same function from the same two numbers, u_h is continuous, $V_h \subset H^1$. §2 proved it from the polynomial; the table proves it from the basis.

So what is *non*-conforming FEA? Giving up $V_h \subset V$ on purpose, and replacing the guarantee with a weaker one that still works.

NOT FROM LECTURE — non-conforming elements and the patch test

The one you have met. *Incompatible modes* — Wilson's Q6 / QM6. Add two internal bubbles to the Q4 field,

$$u = \sum_a N_a u_a + a_1(1 - r^2) + a_2(1 - s^2).$$

On the edge $s = -1$ the second bubble dies but the first does not, so the edge displacement goes *quadratic* in r while the neighbour's a_1 is independent of yours — a gap opens along the shared edge. Deliberately non-conforming, and done because it cures Q4's shear locking in bending: the xy term alone cannot bend without generating parasitic shear.

The other big family: plates. C^1 in 2D costs 21 DOF, so most production plate elements (Morley, DKT, ACM) decline to pay and are non-conforming.

What replaces the guarantee. In engineering, the **patch test**: load a patch of elements with boundary data consistent with a constant strain state, and demand the element reproduce it *exactly*. In the mathematics, **Strang's second lemma**: the error bound picks up an extra *consistency* term measuring how far $B_h(u, v_h) - L(v_h)$ is from zero. They are the same condition. Pass it and you still converge; fail it and the element can converge confidently to the wrong answer — which several early plate elements did.

The three different "conforming"s. This is where the confusion usually comes from — the word is doing three unrelated jobs:

the phrase	what it is about	the condition
conforming element (engineering: "compatible")	the approximation space	$V_h \subset V$
conforming mesh	how elements are laid out	faces meet whole-face to whole-face; no hanging nodes
compatibility (solid mechanics)	a strain field	Saint-Venant: $\boldsymbol{\varepsilon}$ comes from a single-valued $\boldsymbol{u}$

Only the first is what the lecture means. The second is a *meshing* property — you can mesh a perfectly conforming Q4 non-conformingly, with hanging nodes, and lose $V_h \subset V$ that way instead. The third has nothing to do with either.

When is a jump an error, and when is it the answer? **[not from lecture]** Three cases, all three actually solved to draw the figure — nothing in it is sketched by hand.

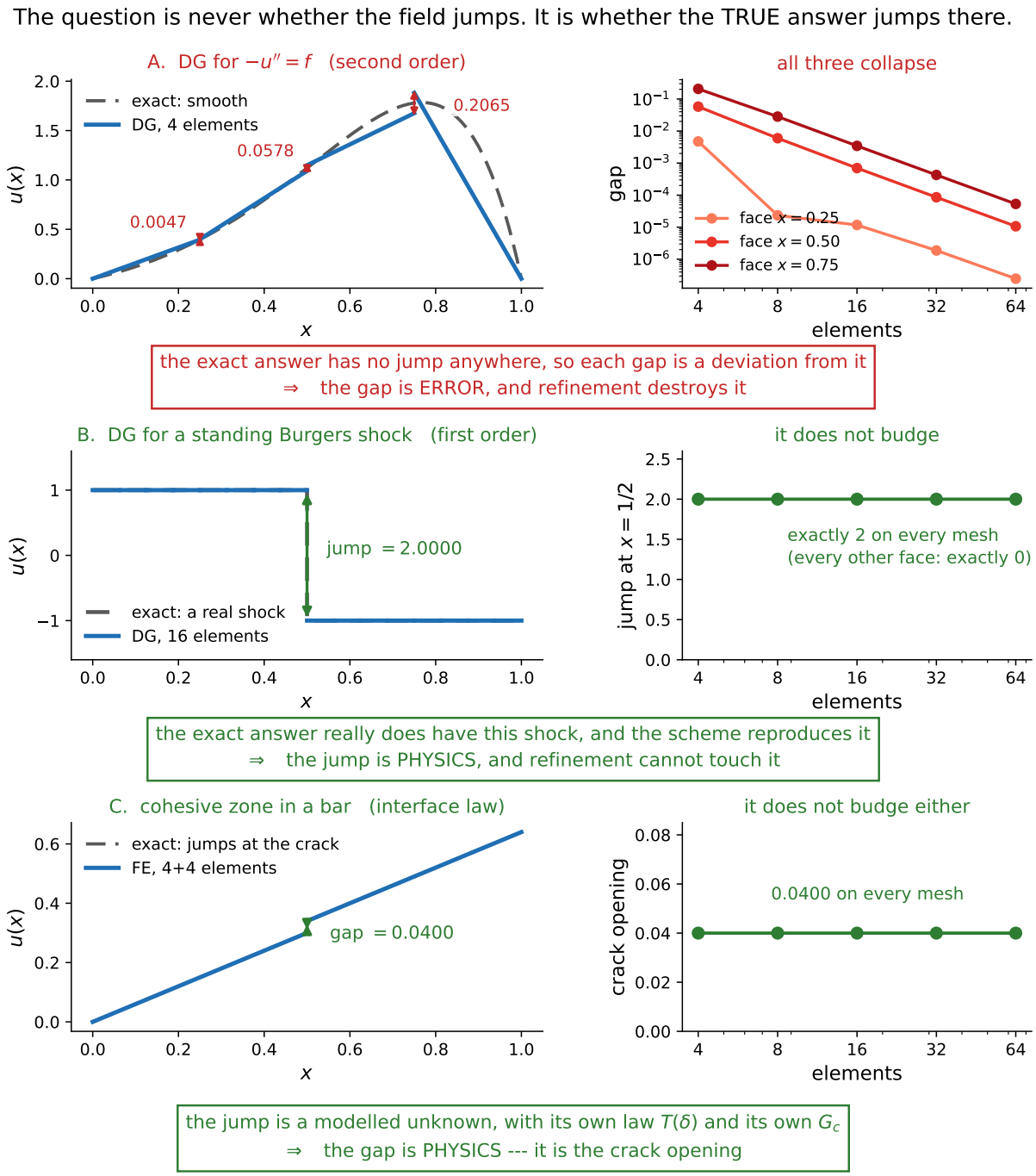


Figure 5: Three fields that all jump at an interface, each one actually solved rather than sketched. **A:** interior-penalty DG on $-u'' = f$, where the exact solution is smooth, so every jump is pure discretisation error and vanishes under refinement. **B:** DG on a stationary shock, where the exact solution really is discontinuous and the jump is the answer, not an error. **C:** a cohesive zone, where the jump is the crack opening and is set by the traction–separation law, not by the mesh. Same picture, three different meanings — which is the whole point.

In all three the field jumps, by construction. What separates them is **whether the *true* answer jumps there**:

	A: DG, $-u'' = f$	B: DG, a shock	C: cohesive zone
nodes doubled	every face	every face	the crack path only
true answer jumps there?	no ×	yes ✓	yes ✓
so the gap is	error	physics	physics
refine the mesh	$\to 0$	stays 2.0000	stays 0.0400

That is a test you can actually run. Refine and watch the gap. If it shrinks, it was discretisation error. If it settles on a number you could measure in a lab, it was physics.

And now the part that trips everyone up: conformity is a *different* question, and it does not line up with that table. Measured against an H^1-type space — the one a second-order equation forces on you — **A and B are *both* non-conforming**: DG breaks the space either way, and column B's upwind flux is a jump penalty too, its size merely fixed by the physics, $\frac{1}{2}|\beta\cdot n|$, instead of tuned. Column **C** is conforming, because the continuum model *itself* admits a jump on the crack plane.

But B's label is not even stable, and that is the real lesson. Write the first-order problem in *flux* form — every derivative moved onto the test function — and the trial space needs no derivative at all, broken polynomials sit inside it, and the same method is *conforming*. Write it in graph form, $\{v : \beta\cdot\nabla v \in L^2\}$, and a jump across a face puts a δ into $\beta\cdot\nabla v$, so it is *non-conforming*. Same method, same mesh, same answer — two labels, because two different V's were written down. So: **never ask "is DG conforming?" Ask "non-conforming with respect to *what*?"** There are only three possible answers, and they live in different sentences:

sense	the claim is about	carries a theorem?
the **space**	$V_h \not\subset V$, with V fixed by *your* weak form	yes — it decides which error analysis applies
the **mesh**	hanging nodes, non-matching faces	no
neither	local slang for "not the standard CG element"	no

Only the first carries any content, and even it is relative. The method's behaviour never changes when the label does — only the yardstick does. **The gap-under-refinement test is worth more than any of the three labels**, because it is a property of your problem rather than of somebody's vocabulary.

Two more traps in the vocabulary while we are here.

- **Non-conforming does not mean inconsistent.** Interior-penalty DG is *consistent*: the exact solution satisfies the discrete equations exactly, so the Strang consistency term is identically zero and Galerkin orthogonality still holds in B_h. Wilson's Q6 and Morley are non-conforming *and* inconsistent — which is exactly why *they* need a patch test and DG does not. The patch test belongs to the classical non-conforming elements, not to DG.
- **"Discrete" is not a third kind of continuity.** Discrete means finite-dimensional — you replaced a PDE by N unknowns. *Every* finite element method is discrete, CG and DG alike. Continuous versus discontinuous is a separate axis: whether the approximation is allowed to jump across a face.

Which sense a book means depends on which building it was written in. **Ern–Guermond** and **Di Pietro–Ern** — this course's dialect — always use the space sense and always *name* the space:

"the approximation setting in dG methods is nonconforming since functions in V_h can jump across the mesh interfaces, whereas membership in $V := H_0^1$ requires continuity". They are not saying DG is non-conforming full stop; they are saying it is non-conforming *against* H_0^1, which is the space their chapter is about. **Computational mechanics** (Zienkiewicz–Taylor and the whole conforming/compatible versus incompatible tradition) uses the blanket sense and settles it with the patch test. And in **CFD and the hyperbolic DG world — the one your MOOSE work sits in** — "non-conforming" most often means the *mesh*: hanging nodes and non-matching faces, nothing to do with function spaces at all.

The intuition: $\det \boldsymbol{J}$ **is a *local* stretch, not a global one.** Picture the reference square as a rubber sheet ruled with a uniform grid. The map pulls it onto D, and because D is not a rectangle the sheet is stretched *unevenly*: a small patch of reference area becomes a larger physical patch in some parts of the element and a smaller one in others. $\det \boldsymbol{J}$ at a point is exactly that local magnification. Four Gauss points sit in four different places, so they see four different magnifications.

You can read it off D's shape. Compare how *open* its corners are — the triangle cut off at each node has area

$$\text{node } 1: \ \tfrac{15}{2}, \qquad \text{node } 2: \ 6, \qquad \text{node } 3: \ \tfrac{13}{2}, \qquad \text{node } 4: \ 8.$$

Node 4 at $(1, 5)$ is the widest corner and node 2 at $(3, 0)$ the tightest, so the map stretches most towards $(r, s) = (-1, 1)$ and least towards $(1, -1)$. The Gauss points follow exactly that: $q4$ (nearest the open corner) carries the *largest* value, 3.789, and $q2$ (nearest the tight one) the *smallest*, 3.211. "Nearest" is not a coincidence: the 2×2 points are ordered $(-g, -g), (g, -g), (g, g), (-g, g)$, the same cycle as the nodes, so q_a always sits in node a's quadrant — before the map and after it.

The check that ties it together. Each Gauss point carries $w_q \det \boldsymbol{J}_q$ of physical area — its own share of the element. Add them up:

$$\sum_{q=1}^{4} w_q \det \boldsymbol{J}_q = 3.644338 + 3.211325 + 3.355662 + 3.788675 = 14.000000,$$

exactly the area of D. So the four numbers are not four competing answers; they are four pieces of one element, and the quadrature weights every integrand value by how much physical area its point stands for. (This sum is exact because $\det \boldsymbol{J}$ is *linear* in r, s — it is the *ratio* in (18) that the rule cannot handle.)

Two questions the picture has to answer — (a) the node pairing.

(a) Must $(x, y) = (0, 0)$ sit at $(r, s) = (-1, -1)$? No — that is this element's node numbering, nothing more. Whichever corner you choose to call node 1 goes to $(-1, -1)$; node 2 to $(1, -1)$, node 3 to $(1, 1)$, node 4 to $(-1, 1)$.

The table in the figure box above is the whole of it: at a parent corner three of the four N_a vanish and the fourth is 1, so $\sum_a N_a x_a$ hands back *whichever* nodal coordinate you happened to put in that slot. **Nothing in that calculation knows which physical corner is which** — it only reads the list (x_a, y_a) you supplied. Start D's corner list at a different corner, still going counter-clockwise, and you get the *same element*, relabelled:

list starts at	$\det \boldsymbol{J}(r, s)$	at nodes $1, 2, 3, 4$
$(0, 0)$	$\frac{7}{2} - \frac{3r}{8} + \frac{s}{8}$	$\frac{15}{4},\ 3,\ \frac{13}{4},\ 4$
$(3, 0)$	$\frac{7}{2} + \frac{r}{8} + \frac{3s}{8}$	$3,\ \frac{13}{4},\ 4,\ \frac{15}{4}$
$(4, 4)$	$\frac{7}{2} + \frac{3r}{8} - \frac{s}{8}$	$\frac{13}{4},\ 4,\ \frac{15}{4},\ 3$
$(1, 5)$	$\frac{7}{2} - \frac{r}{8} - \frac{3s}{8}$	$4,\ \frac{15}{4},\ 3,\ \frac{13}{4}$

The same four numbers every time, just rotated around; the area is 14 in all four. What is *not* free is the direction you go round. List D's corners *clockwise* and $\det \boldsymbol{J} = \frac{3r}{8} + \frac{s}{8} - \frac{7}{2}$, giving $-4, -\frac{13}{4}, -3, -\frac{15}{4}$ × — all four negative. The two kinds of swap are not alike, and it is worth knowing which is which: *opposite* nodes ($1 \leftrightarrow 3$ or $2 \leftrightarrow 4$) merely reverses the traversal, so you land back on the clockwise case $(-\frac{13}{4}, -3, -\frac{15}{4}, -4)$ and the quadrilateral stays simple — which is exactly why that swap is the *cure* for a clockwise list. *Adjacent* nodes ($2 \leftrightarrow 3$, say) is the damaging one: it gives $4, -3, -\frac{13}{4}, \frac{15}{4}$ ×, mixed signs, the bow-tie. So **which** corner you start at is a free choice; **the cyclic order is not** — and it is not a per-element whim either, since the element across a shared edge must list that edge's two nodes the same way or the two images of the edge will not coincide.

(b) Why does the grid run parallel to D's edges instead of staying square? Because the edges *are* grid lines. They are not a separate object drawn on top of the grid — they are its outermost members:

$$r = -1 \text{ is edge } 1{-}4, \quad r = +1 \text{ is edge } 2{-}3, \quad s = -1 \text{ is edge } 1{-}2, \quad s = +1 \text{ is edge } 4{-}3.$$

Every line $r = c$ runs from edge $1{-}2$ to edge $4{-}3$ and interpolates between edges $1{-}4$ and $2{-}3$ on the way, so of course it looks like them. It does *not* stay parallel to the r and s axes, because the map is not the identity — it rotates, stretches and shears. Only for a rectangle with edges along x and y would the image grid come out axis-parallel.

And the lines are not parallel to *one another* either. Each image is still perfectly *straight* — along $r = c$ both x and y are affine in s — but the family fans:

line	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	$+1$
slope of $r = \text{const}$	5	$\frac{19}{4}$	$\frac{9}{2}$	$\frac{17}{4}$	4
slope of $s = \text{const}$	0	$-\frac{1}{12}$	$-\frac{1}{6}$	$-\frac{1}{4}$	$-\frac{1}{3}$

The endpoints of each row are just the slopes of the two opposite edges of D — 5 and 4 for edges $1{-}4$ and $2{-}3$, 0 and $-\frac{1}{3}$ for edges $1{-}2$ and $4{-}3$ — and the interior lines rotate smoothly from one to the other. **The fan is the distortion.** Make D a parallelogram and opposite edges become parallel, each row collapses to a single slope, the image grid becomes a true parallelogram grid — and that is the *same* condition that makes $\det \boldsymbol{J}$ constant.

Is $\det \boldsymbol{J}$ the same at the four parent corners? It depends on the element. Put all of them through it, together with a fifth, E, a 10×10 square with corners $(0,0), (-10,0), (-10,-10), (0,-10)$:

element	shape	$\det \boldsymbol{J}(r,s)$	at nodes $1,2,3,4$
A	square 1×1	$\frac{1}{4}$	$\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}$
B	rectangle 2×2	1	$1,\ 1,\ 1,\ 1$
C	rectangle 3×1	$\frac{3}{4}$	$\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}$
E	square 10×10	25	$25,\ 25,\ 25,\ 25$
D	general convex quad	$\frac{7}{2} - \frac{3r}{8} + \frac{s}{8}$	$\frac{15}{4},\ 3,\ \frac{13}{4},\ 4$

A, B, C, E all give four *equal* values; only D does not. The case where they are equal has a name.

(i) Constant $\det \boldsymbol{J} \iff$ the element is a parallelogram. The rs coefficient of the map is

$$x(r,s) \text{ has } rs\text{-coefficient } \tfrac{1}{4}(x_1 - x_2 + x_3 - x_4), \qquad y(r,s) \text{ has } \tfrac{1}{4}(y_1 - y_2 + y_3 - y_4),$$

and both vanish exactly when $x_3 = x_2 + x_4 - x_1$, $y_3 = y_2 + y_4 - y_1$ — the definition of a parallelogram. With no rs term the map is affine, $\boldsymbol{J}$ is a constant matrix, and every corner gives the same $\det \boldsymbol{J}$. A rectangle is the special case. For D: $y_1 - y_2 + y_3 - y_4 = 0 - 0 + 4 - 5 = -1 \neq 0$ × — not a parallelogram, so $\det \boldsymbol{J}$ must vary. (Its x counterpart is $0 - 3 + 4 - 1 = 0$, which is why $x(r,s)$ in (25) has no rs term while $y(r,s)$ does.)

(ii) $\det \boldsymbol{J}$ is always *affine* in (r,s) — never bilinear. Working it out for a quad with arbitrary corners, the rs term of $\det \boldsymbol{J}$ *cancels identically*:

$$\det \boldsymbol{J} = \alpha + \beta r + \gamma s \qquad (\text{no } rs \text{ term, for any Q4}).$$

That has a practical consequence: an affine function on a square attains its maximum and minimum **at the corners**. So testing $\det \boldsymbol{J} > 0$ at the four corners certifies it is positive *everywhere* in the element — no interior check needed. This is exactly why a mesher's element-quality routine reports *corner* Jacobians.

(iii) The four corner values average to the centre value. Again because $\det \boldsymbol{J}$ is affine,

$$\tfrac{1}{4}\big(3.75 + 3 + 3.25 + 4\big) = 3.5 = \det \boldsymbol{J}\big|_{(0,0)}, \qquad 3.5 \times \underbrace{4}_{\text{reference area}} = 14 = \text{area of } D.$$

So the corner Jacobians are not four unrelated numbers: their mean is the element's average magnification, and that times the reference area is the area.

(iv) Two things element E makes visible. First, *scale*: E has $100\times$ the area of A and exactly $100\times$ its $\det \boldsymbol{J}$ (25 against $\frac{1}{4}$). $\det \boldsymbol{J}$ is an area magnification, so it grows with the square of the element size — halving a mesh divides every $\det \boldsymbol{J}$ by four.

Second, a trap. E's map is

$$x(r,s) = -5r - 5, \qquad y(r,s) = -5s - 5, \qquad \text{so} \qquad \boldsymbol{J} = \begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}, \qquad \det \boldsymbol{J} = (-5)(-5) - 0 = +25.$$

Both entries of $\boldsymbol{J}$ are negative — x *decreases* as r increases, and likewise y with s — yet $\det \boldsymbol{J} > 0$ and the element is perfectly valid. Flipping *two* axes is a 180° rotation, not a reflection, so orientation is preserved. The lesson: **never read the signs of the individual entries of $\boldsymbol{J}$; read $\det \boldsymbol{J}$.** Only an *odd* number of flips inverts an element.

Caveat. All three statements are specific to the *bilinear* Q4. For a quadratic element (Q8/Q9) or a curved edge, $\det \boldsymbol{J}$ is no longer affine, its extrema can sit in the interior, and checking only the corners is *not* sufficient.

Four things about $\det \boldsymbol{J}$ worth carrying around.

1. A corner Jacobian *is* a triangle area. At node i, let $\boldsymbol{v}_{\text{next}}$ and $\boldsymbol{v}_{\text{prev}}$ be the two edge vectors leaving it. Then

$$\det \boldsymbol{J}\big|_{\text{node } i} = \tfrac{1}{4}\big(\boldsymbol{v}_{\text{next}} \times \boldsymbol{v}_{\text{prev}}\big) = \tfrac{1}{2} \times \text{area of the triangle cut off at that corner.}$$

For D at node 1: neighbours $(3,0)$ and $(1,5)$, triangle area $\frac{15}{2}$, half of it $\frac{15}{4}$ — exactly the $\det \boldsymbol{J}$ tabulated above. *Use:* you can estimate corner Jacobians *by eye* — the more open the corner, the larger its Jacobian. A nearly-straight corner has one near zero.

2. All four negative ⇒ you numbered the nodes clockwise. Numbering D backwards gives $-\frac{15}{4}, -4, -\frac{13}{4}, -3$ — same magnitudes, every sign flipped. The element is fine; the *connectivity* is reversed. Fix by swapping two *opposite* nodes ($1 \leftrightarrow 3$ or $2 \leftrightarrow 4$), not by remeshing — an *adjacent* swap would tangle it instead.

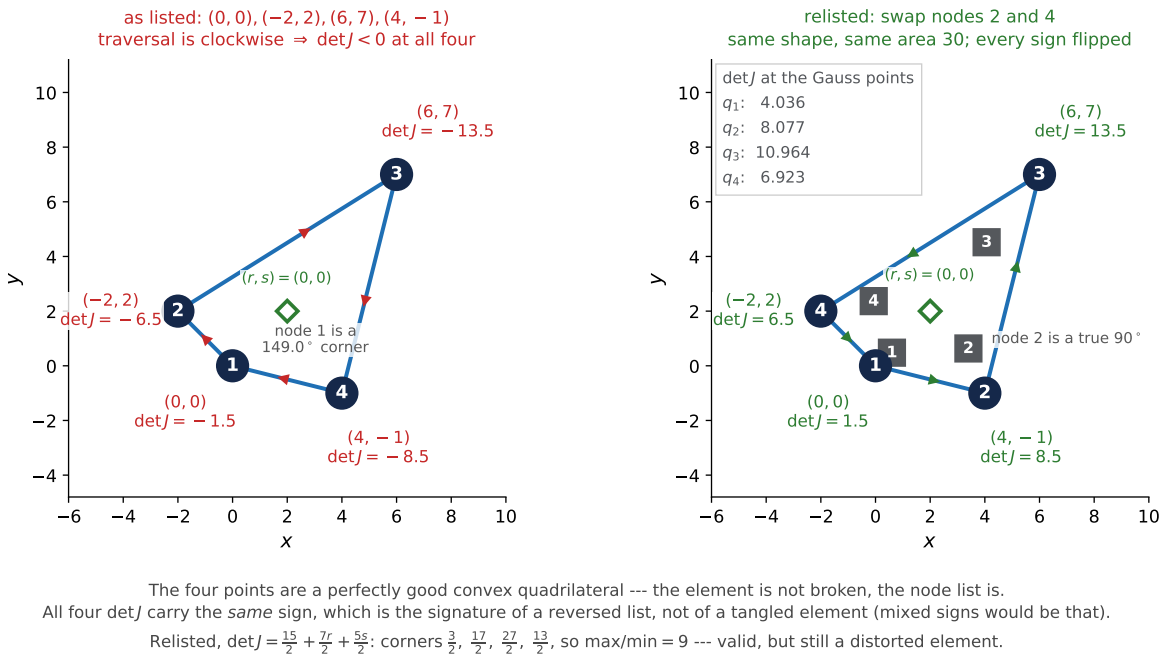
3. *One* negative ⇒ the element is genuinely tangled. A quad with a reflex (concave) corner at node 3 gives $2,\ \frac{1}{4},\ -\frac{3}{2},\ \frac{1}{4}$ — and the negative one *is* the reflex corner. Mixed signs cannot be cured by renumbering. So the sign pattern diagnoses the problem: **uniform sign = orientation, mixed sign = geometry**.

4. $\det \boldsymbol{J}$ tells you nothing about skew. Shear a unit parallelogram by k — corners $(0,0), (1,0), (1{+}k, 1), (k, 1)$:

k	0	1	3	10	50
$\det \boldsymbol{J}$	1/4	1/4	1/4	1/4	1/4
area	1	1	1	1	1
$\text{cond}(\boldsymbol{J})$	1.0	2.6	10.9	102	2502

The determinant and the area never move, while the element becomes a worse and worse sliver. **A healthy $\det \boldsymbol{J}$ does not mean a healthy element.** That is why mesh-quality reports carry aspect ratio and skew *separately* from the Jacobian — and why Lecture 13's regularity condition $h_j/\rho_j \le \alpha$ is a statement about *shape*, not about area.

A worked case: element F, listed the wrong way round. Take $(0,0)$, $(-2,2)$, $(6,7)$, $(4,-1)$ as nodes $1,2,3,4$ and run the whole diagnosis above on it — Figure 6.



The four points are a perfectly good convex quadrilateral --- the element is not broken, the node list is.
All four det$J$ carry the *same* sign, which is the signature of a reversed list, not of a tangled element (mixed signs would be that).
Relisted, $\det J = \frac{15}{2} + \frac{7r}{2} + \frac{5s}{2}$: corners $\frac{3}{2}$, $\frac{17}{2}$, $\frac{27}{2}$, $\frac{13}{2}$, so max/min = 9 --- valid, but still a distorted element.

Figure 6: Element F, the same four points listed two ways. **Left:** as given, which traverses them clockwise — $\det \boldsymbol{J} < 0$ at every Gauss point. **Right:** relisted counter-clockwise, same geometry, $\det \boldsymbol{J} > 0$ throughout. The shape was never the problem; the connectivity order was.

The shape is fine. No edge crosses another, and all four corner cross products share a sign, so it is a simple *convex* quadrilateral. Interior angles $149.0°, 77.0°, 44.0°, 90.0°$ (sum 360° ✓), area 30.

The list is not. As given,

$$x = 2 + 3s + rs, \qquad y = 2 + \tfrac{5r}{2} + s + \tfrac{3rs}{2}, \qquad \det \boldsymbol{J} = -\tfrac{15}{2} - \tfrac{5r}{2} - \tfrac{7s}{2},$$

and at nodes $1, 2, 3, 4$ that reads $-\frac{3}{2}$, $-\frac{13}{2}$, $-\frac{27}{2}$, $-\frac{17}{2}$ × — four negatives. By point 2 a *uniform* sign is an orientation report, not a geometry report, and the shoelace sum confirms it: $2A = -60 < 0$, so the corners were walked **clockwise**.

The repair is the one point 2 prescribes — swap two *opposite* nodes, here $2 \leftrightarrow 4$, giving $1{=}(0,0)$, $2{=}(4,-1)$, $3{=}(6,7)$, $4{=}(-2,2)$:

$$x = 2 + 3r + rs, \qquad y = 2 + r + \tfrac{5s}{2} + \tfrac{3rs}{2}, \qquad \det \boldsymbol{J} = \tfrac{15}{2} + \tfrac{7r}{2} + \tfrac{5s}{2},$$

with nodal values $\frac{3}{2}$, $\frac{17}{2}$, $\frac{27}{2}$, $\frac{13}{2}$ ✓— the same four magnitudes, every sign flipped, exactly as point 2 says. The quadrature closes too: $4.035898 + 8.077350 + 10.964102 + 6.922650 = 30$, the area. **Nothing was remeshed.** One transposition in the connectivity array.

It is a valid element and a mediocre one, and these are separate verdicts. Its corner Jacobians span $\max / \min = \frac{27/2}{3/2} = 9$. Against the last box's scale — 1 for any parallelogram — that is still poor. Point 1 says where it comes from: node 1 is a 149.0° corner, so the triangle cut off there has area only 3 and half of it is $\det \boldsymbol{J} = \frac{3}{2}$, while node 3's triangle has area 27 and gives $\frac{27}{2}$. The element would pass every validity check in this note and still integrate indifferently, because $\det \boldsymbol{J}$ in the denominator of (18) swings by a factor of 9 across it.

(It is not a parallelogram, which is why $\det \boldsymbol{J}$ varies at all: the rs coefficients are $\frac{1}{4}(0 - 4 + 6 + 2) = 1$ and $\frac{1}{4}(0 + 1 + 7 - 2) = \frac{3}{2}$, neither zero, so test (i) fails. Two things it *does* have: the corner at $(4, -1)$ is a true right angle, since $(4, -1) \cdot (2, 8) = 0$, and the edge beyond it is exactly twice as long, $\sqrt{17}$ against $2\sqrt{17}$. Neither helps — $\det \boldsymbol{J}$ does not read angles, only areas.)

Geometry as a variable — what moves $\det \boldsymbol{J}$, and what does not. Each column below starts from the *same* unit square, where $\det \boldsymbol{J} = \frac{1}{4}$ at all four nodes and at all four Gauss points, and then changes exactly one number. The top row is the geometry; the bottom row is $\det \boldsymbol{J}$ at the four corner nodes (coloured) and at the four Gauss points (grey) as that number is swept.

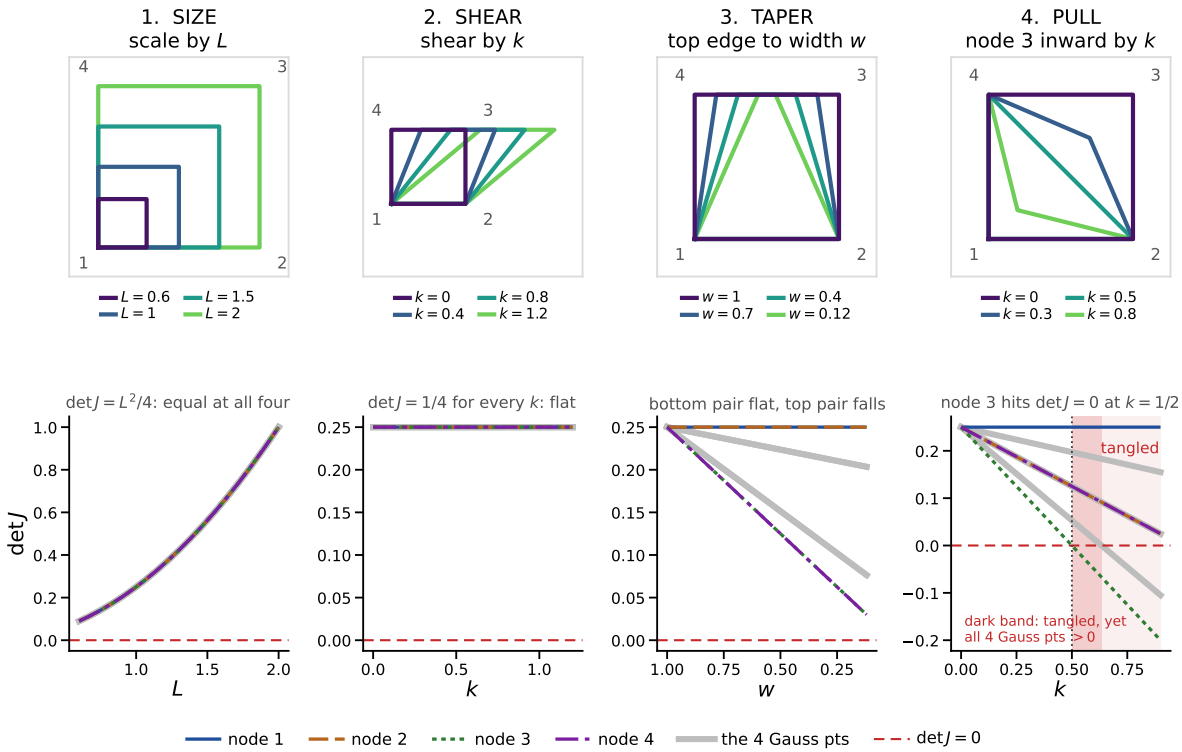


Figure 7: Geometry as a variable: four one-parameter families, with $\det \boldsymbol{J}$ computed from the nodal coordinates at the four corner nodes (coloured) and the four Gauss points (grey) as the parameter is swept. SIZE and SHEAR leave $\det \boldsymbol{J}$ constant — both keep the element a parallelogram. TAPER and PULL do not, and the spread between largest and smallest is the measure of how far from a parallelogram the element has drifted.

family	$\det \boldsymbol{J}(r,s)$	at nodes $1,2,3,4$	max / min
SIZE, L	$L^2/4$	all four $= L^2/4$	1
SHEAR, k	$1/4$	all four $= 1/4$	1
TAPER, w	$\big[(1+w)+(w-1)s\big]/8$	$\frac{1}{4},\ \frac{1}{4},\ \frac{w}{4},\ \frac{w}{4}$	$1/w$
PULL, k	$\big[2-2k-k(r+s)\big]/8$	$\frac{1}{4},\ \frac{1-k}{4},\ \frac{1-2k}{4},\ \frac{1-k}{4}$	$1/(1-2k)$

1. SIZE — $\det \boldsymbol{J}$ moves a lot; the element does not get worse. Scaling multiplies every $\det \boldsymbol{J}$ by L^2 and leaves all four equal. The curve is a parabola, the four nodes sit on top of one another, and the Gauss points sit on them too. Halving a mesh divides every $\det \boldsymbol{J}$ by four — a large change in the *number* with no change at all in element *quality*.

2. SHEAR — $\det \boldsymbol{J}$ does not move at all. A dead-flat line at $\frac{1}{4}$ while the element slides into a sliver. This is point 4 of the previous box drawn as a graph: the determinant is an area, and shearing preserves area. **A flat $\det \boldsymbol{J}$ curve is not a certificate of a good element.**

Geometry as a variable, continued — columns 3 and 4.

3. TAPER — the four values split into two pairs. $\det \boldsymbol{J}$ depends on s alone, so the bottom nodes $1, 2$ hold $\frac{1}{4}$ for every w while the top nodes $3, 4$ fall to $\frac{w}{4}$. This is the first case where "the Jacobian of the element" stops being a single number: at $w = 0.12$ the corner values are $\frac{1}{4}, \frac{1}{4}, 0.03, 0.03$ — a factor of 8.3 across one element. Notice the grey curves lie strictly *between* the coloured ones: the Gauss points are interior, so they never attain the extreme values.

4. PULL — a corner runs out. Dragging node 3 in along the diagonal makes $\det \boldsymbol{J}$ depend on both r and s. Node 1 is untouched, nodes 2 and 4 fall together, and node 3 falls twice as fast and reaches *zero* at $k = \frac{1}{2}$. That is exactly where node 3 lands on the straight line joining nodes 2 and 4: by point 1 of the previous box the corner Jacobian *is* half the area of the triangle cut off at that corner, and at $k = \frac{1}{2}$ that triangle has collapsed. Past $k = \frac{1}{2}$ the sign flips — one negative corner, the mixed-sign pattern of a genuinely tangled element.

The trap the fourth column was drawn for. In the dark band, $\frac{1}{2} < k < \frac{3-\sqrt{3}}{2} \approx 0.634$, the element is already tangled — node 3 has $\det \boldsymbol{J} < 0$ — and yet **all four 2×2 Gauss points still return** $\det \boldsymbol{J} > 0$. At $k = 0.6$ the corners read $0.25,\ 0.10,\ -0.05,\ 0.10$ while the Gauss points read $0.187,\ 0.100,\ 0.013,\ 0.100$. A solver that tests only its quadrature points accepts the element, assembles $\boldsymbol{k}^e$ from it, and reports nothing. The reason is the affineness established two boxes back, $\det \boldsymbol{J} = \alpha + \beta r + \gamma s$: its extremes sit at the *corners*, while the Gauss points sit strictly inside at $\pm 1/\sqrt{3}$ and can only ever report values in between. **Test the corners, not the Gauss points.**

What to carry away. $\det \boldsymbol{J}$ itself tells you size (column 1) and validity (column 4). It tells you nothing about skew (column 2). The number that tells you whether the mapping is *distorted* is the *spread* of $\det \boldsymbol{J}$ across the element — the last column of the table — which is 1 for any parallelogram and grows without bound as a corner closes up. That spread is the computational face of Lecture 13's shape-regularity condition.

A Gauss rule is constructed to integrate *polynomials* exactly — that is the whole content of "exact through degree $2n - 1$". The right-hand side of (18) is not a polynomial: no finite-order Gauss rule integrates it exactly, only ever approximately, and the approximation improves as points are added. For D:

rule	2×2	3×3	6×6	converged
k^e_{11}	0.675085	0.675272	0.675272	0.675272

— an error of about 0.03% at 2×2, settled by about three points per direction. In practice 2×2 is used for Q4 anyway and that small quadrature error is accepted; the point here is only that "exact" is a claim about rectangles, not about every element.

Had we stayed in x, y there would be no table to read from. The integration limits would be the element's own edges, different for every element — and for a quad like D not even expressible as constant limits — so the abscissae and weights would have to be rebuilt element by element. Mapping to $[-1, 1]^2$ is what turns integration into a fixed, four-line loop.

The check that settles it. Move the evaluation point to $(r, s) = (1, 1)$, which is node 3 itself. There $N_3 = 1$ and $N_1 = N_2 = N_4 = 0$: node 3 takes the whole share. In A, B and C this can be confirmed from the x, y formula — it gives 1 at $(1, 1)$, $(3, 2)$ and $(5, 4)$ respectively. **In D there is no formula to confirm it with, and yet the value is still exactly** 1 — because the nodal condition (27) is imposed in (r, s), where every element is the same square. We know the answer without being able to write the function. That is the whole case for the reference square in one line.

			at the interior point $(r,s)=(0.5,0.5)$		at node 3 itself, $(r,s)=(1,1)$	
element	region	N_3 in x,y	sits at $(x,y)=$	node 3's share	sits at $(x,y)=$	node 3's share
A	$[0,1]\times[0,1]$	$x\,y$	(0.75, 0.75)	0.5625	(1, 1)	**1**
B	$[1,3]\times[0,2]$	$\frac{1}{4}(x-1)\,y$	(2.50, 1.50)	0.5625	(3, 2)	**1**
C	$[2,5]\times[3,4]$	$\frac{1}{3}(x-2)(y-3)$	(4.25, 3.75)	0.5625	(5, 4)	**1**
D	a general quad	none	(3, 3.1875)	0.5625	(4, 4)	**1**
any	—	$\frac{1}{4}(1+r)(1+s)$	$(r,s)=(0.5,0.5)$	**0.5625**	$(r,s)=(1,1)$	**1**

Table 7: The contribution of *node* 3, on four elements. Columns 4–5 read it at a general interior point, columns 6–7 at node 3 itself. The "sits at" columns differ element by element and the x,y formula differs too — but the two share columns do not. The last row is the only one a program needs.

Row D is the one that settles it. in a arbitrary physical x-y co-ordinate mesh– Given the quad the corners $(0,0)$, $(3,0)$, $(4,4)$, $(1,5)$, taken counter-clockwise from local node 1. There is no formula in x,y to evaluate — the "N_3 in x,y for local node 3" column is empty. The physical point is found instead from the *isoparametric* map, defined next.

What "isoparametric" means

The name decodes directly. *Iso* = same, *parametric* = the shape functions. An element is **isoparametric** when the *same* shape functions are used for two different jobs:

job	what is interpolated	written out
interpolate the **unknown**	the nodal values u_a	$u_h\big\vert_K=\sum_{a=1}^{4}N_a(r,s)\,u_a$
interpolate the **geometry**	the nodal coordinates (x_a,y_a)	$x=\sum_{a=1}^{4}N_a(r,s)\,x_a,\quad y=\sum_{a=1}^{4}N_a(r,s)\,y_a$

Same N_a in both rows. That — and only that — is the definition. It also names the two alternatives, which is the distinction to have ready:

	geometry order	typical use
subparametric	*lower* than the solution	straight-sided element carrying a quadratic solution
isoparametric	the *same* as the solution	the standard choice; curved edges when the element is curved
superparametric	*higher* than the solution	rare

Why it is correct, not merely convenient. Take any linear field $u=\alpha+\beta x+\gamma y$, sample it at the four corners, and interpolate:

$$u_h=\sum_a N_a u_a=\alpha\underbrace{\sum_a N_a}_{=1\text{ by (36)}}+\beta\underbrace{\sum_a N_a x_a}_{=x\text{ by the map}}+\gamma\underbrace{\sum_a N_a y_a}_{=y\text{ by the map}}=\alpha+\beta x+\gamma y=u. \tag{19}$$

The second and third underbraces are true *because the map is built from the same* N_a used in geometry interpolation. Choose a different set of N for geometry interpolation and than solution interpolation

fails . So the element passes the patch test by construction, on a distorted quad as much as on a square regardless of the element shape.

Checked on element D with $u = 2 + 3x - y$ (nodal values $2, 11, 10, 0$): at $(r, s) = (0.5, 0.5)$, $(-0.75, 0.333)$, $(0, 0)$ and $(0.9, -0.875)$ the interpolated value equals the exact value of u at the mapped point, to the last digit of exact arithmetic.

What the three u's are. (19) carries three symbols that all look like "u". They are three different *kinds* of object, and the lectures use all three:

symbol	what kind of thing it is	lecture's symbol	on element D
u	a **function**, defined at every point of the domain	u	$u = 2 + 3x - y$, the field you are testing with
u_a	**numbers** — one per node, four of them	ξ_j, the coefficients	$u_1, \dots, u_4 = 2,\ 11,\ 10,\ 0$
u_h	a **function** again, but only one of those this mesh is able to build	$u_h \in V_h$	$u_h = \sum_a N_a u_a$

Table 8: The three objects (19) juggles. Two of them are functions and one is a list of numbers — which is why they cannot share a symbol.

Why one symbol will not do. Because in general $u \neq u_h$, and the gap between them *is* the subject of the error analysis — Céa's lemma bounds $\|u - u_h\|_V$, and you cannot write that quantity down if the exact field and the computed field share a name. And u_a is not a function at all: it is the list of numbers the solver stores and solves for, which N_a turns back into a function. In the lecture's symbols the same three are u, ξ_j, and $u_h = \sum_j \xi_j \varphi_j$ — the only translation needed is $N_a \leftrightarrow \varphi_j$ and $u_a \leftrightarrow \xi_j$.

Reading (19) one step at a time. The question it answers is: *hand the element a flat tilted plane — does it give the same plane back?*

1. **The element only ever receives four numbers.** Visit each node, read its coordinates, evaluate the plane there: $u_a = \alpha + \beta x_a + \gamma y_a$. On element D with $u = 2 + 3x - y$ those are $2, 11, 10, 0$. The formula itself is then thrown away — the element never sees it.

2. **Stand at an interior point and ask two *separate* questions.** At $(r, s) = (0, 0)$ every $N_a = \frac{1}{4}$, so

$$\underbrace{x = \textstyle\sum_a N_a x_a = \frac{1}{4}(0 + 3 + 4 + 1) = 2.00}_{\text{"where am I?"}} \qquad \underbrace{u_h = \textstyle\sum_a N_a u_a = \frac{1}{4}(2 + 11 + 10 + 0) = 5.75}_{\text{"what is the field here?"}}$$

   and likewise $y = \frac{1}{4}(0 + 0 + 4 + 5) = 2.25$.

3. **Check it.** The true field at $(2.00,\ 2.25)$ is $2 + 3(2.00) - 2.25 = 5.75$. Equal, to the last digit.

4. **Now do step 3 symbolically instead of numerically** — that is all (19) is. Substitute $u_a = \alpha + \beta x_a + \gamma y_a$ into $u_h = \sum_a N_a u_a$, then pull the constants out of the sum, which is legal because α, β, γ are the same at every node:

$$u_h = \alpha\Big(\sum_a N_a\Big) + \beta\Big(\sum_a N_a x_a\Big) + \gamma\Big(\sum_a N_a y_a\Big).$$

   One substitution and one distribution. Nothing else happens. What remains is to name the three sums:

the sum	equals	why	its share at $(r,s)=(0,0)$
$\sum_a N_a$	1	partition of unity, (36)	$\alpha \cdot 1 = 2.00$
$\sum_a N_a x_a$	x	this *is* the map — it is the formula that told you where you were standing	$\beta \cdot 2.00 = 6.00$
$\sum_a N_a y_a$	y	likewise	$\gamma \cdot 2.25 = -2.25$
		so $u_h = \alpha + \beta x + \gamma y = u$, and numerically	5.75

Table 9: The three sums (19) breaks into, and what each one is. The middle row is the only one that needs the element to be isoparametric.

The step that looks circular, and is not. Writing $\sum_a N_a x_a = x$ feels like assuming the answer. Look back at step 2: the number 2.00 appeared *twice* — once as the reply to "where am I?" and once inside the field calculation. Same formula, same number. That happens only because both questions are answered with the same N_a. Give the geometry a different set of functions and "where am I?" returns one number while $\sum_a N_a x_a$ returns another; the three sums stop collapsing and (19) fails. That is the entire cash value of the word *isoparametric*.

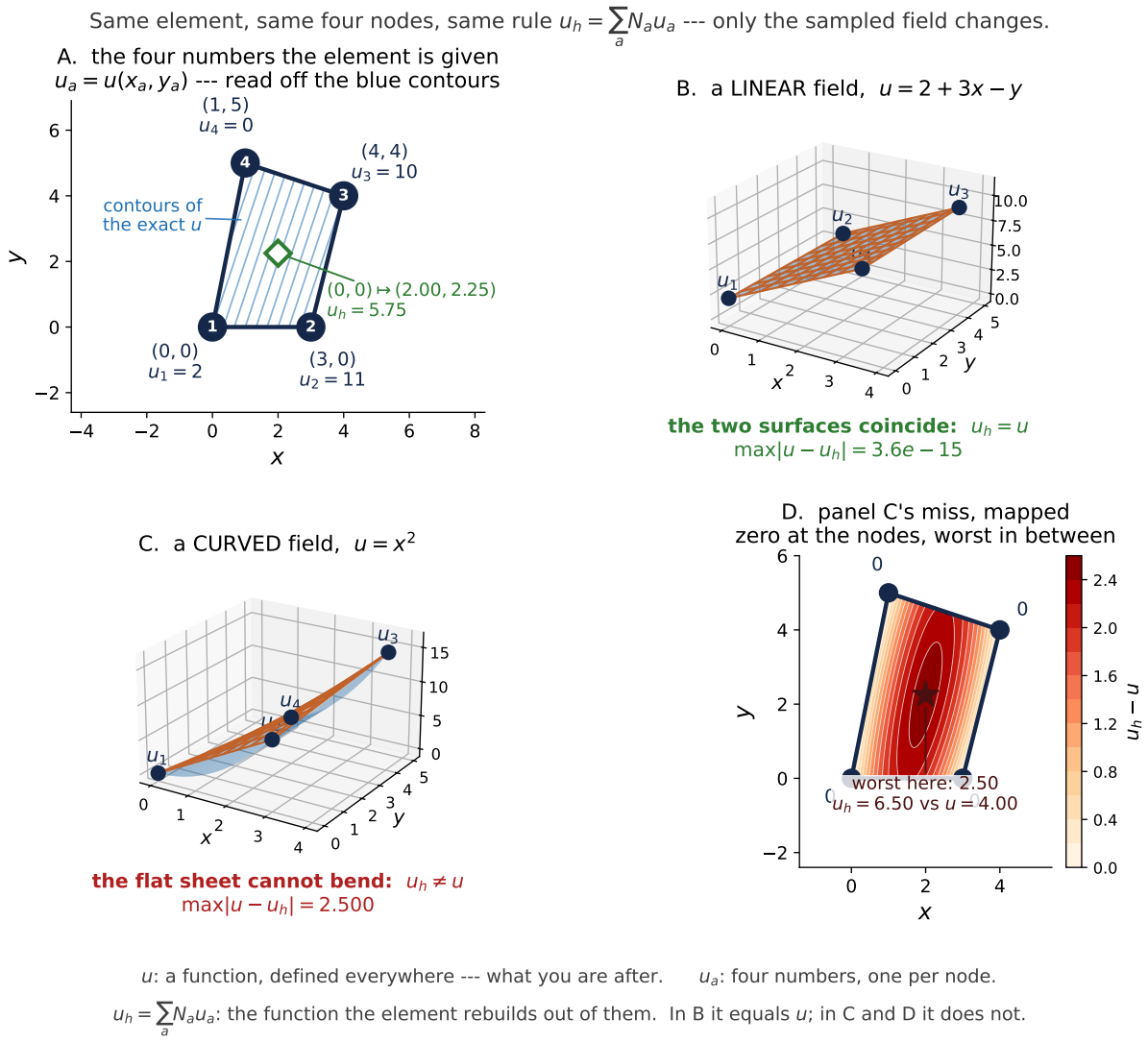


Figure 8: A: where the four numbers come from — the blue contours are the exact field u, and u_a is just its value read off at node a. Green: the centre $(r, s) = (0, 0)$, its image, and u_h there. **B:** the linear field of (19). The blue surface is u, the orange wireframe is u_h, the dots are the four u_a — and the two surfaces are the *same* surface, $\max|u - u_h| = 3.6 \times 10^{-15}$. There is nothing to see, which is the point. **C:** the identical element and the identical rule, but sampling $u = x^2$. The four dots still sit on u by construction, yet in between the bilinear sheet cannot curve. **D:** that miss, mapped over the element — exactly zero at the four nodes, because the element was *told* those values, and worst in between: $\max|u - u_h| = 2.500$ at the centre, where $u_h = 6.50$ against $u = 4.00$. Panels C and D are why the subscript h has to exist at all. Note that the element shape is the same in all four: distortion is not what breaks C — the curvature of u is.

NOT FROM LECTURE — the six modes of a linear displacement field — which term is which

Equation (19) used a *scalar* linear field. In 2D elasticity the displacement has two components, so the linear field carries *six* constants — and each one is named by the term it multiplies:

$$u = \underbrace{\alpha_1}_{\substack{\text{slide in } x \\ \text{TRANSLATION}}} + \underbrace{\beta_1 x}_{\substack{\text{stretch in } x \\ \varepsilon_{xx}}} + \underbrace{\gamma_1 y}_{\substack{\text{tilts the} \\ \text{VERTICAL edge}}}$$

$$v = \underbrace{\alpha_2}_{\substack{\text{slide in } y \\ \text{TRANSLATION}}} + \underbrace{\beta_2 x}_{\substack{\text{tilts the} \\ \text{HORIZONTAL edge}}} + \underbrace{\gamma_2 y}_{\substack{\text{stretch in } y \\ \varepsilon_{yy}}}$$

Four of the six read off directly. α_1, α_2 contain no x or y, so every point of the element gets the same push: nothing deforms, it just slides. The *diagonal* pair $\beta_1 = \partial u/\partial x$ and $\gamma_2 = \partial v/\partial y$ are the two stretches — walk along x and the x-displacement grows, so the element lengthens in x.

The other two are the interesting ones. The *off-diagonal* pair $\gamma_1 = \partial u/\partial y$ and $\beta_2 = \partial v/\partial x$ is the cross-coupling: move in y and get pushed sideways, move in x and get pushed up. Each one tilts a single edge of the element, and *both* rotation and shear live in that pair — neither γ_1 nor β_2 is one or the other by itself.

NOT FROM LECTURE — rotation vs. shear — what the two tilts do together

Which of the two you have depends entirely on how the tilts sit relative to each other:

γ_1, β_2 lean …	so the element …	and the pair is …
the *same* rotational way	spins; the corner stays 90°	ROTATION
into each other	its 90° corner closes up	SHEAR

$$\underbrace{\gamma_1 + \beta_2}_{=\gamma_{xy}\ \text{(shear)}} \qquad \underbrace{\beta_2 - \gamma_1}_{=2\omega\ \text{(rotation)}}$$

The *sum* of the two tilts is the deformation; their *difference* is the spin. The one test that separates them: *did the right angle survive?* Equivalently, the displacement gradient splits once and for all into a symmetric part and a skew part, $\nabla \boldsymbol{u} = \boldsymbol{\varepsilon} + \boldsymbol{W}$, and those are exactly strain and rotation.

The count. 2 translation + 1 rotation + 3 constant strain = 6 — precisely the six constants above, nothing left over and nothing double-counted. The first three store no strain energy, which is why a correctly formed k^e has exactly three zero eigenvalues in 2D; the last three are what the patch test demands the element reproduce exactly, and what (19) guarantees it does.

The one thing that can go wrong. The map must be invertible, which needs $J > 0$ *everywhere* in the element, not just at the corners. Distort a quad far enough — make it concave, or let two corners cross — and J changes sign, the element folds over itself, and the computed stiffness is meaningless. This is precisely what a mesher's "negative Jacobian" error reports, and it is why element-quality checks exist.

Say it in one sentence. "Isoparametric means the element uses the same shape functions to describe its own geometry as it does to interpolate the solution."

Step (i) — the four shape functions at $(r, s) = (0.5, 0.5)$. From (34), substituting $r = 0.5$ and

$s = 0.5$:

$$\begin{aligned} N_1 = \tfrac{1}{4}(1-r)(1-s) &= \tfrac{1}{4}(0.5)(0.5) &= \tfrac{1}{16} &= 0.0625, \\ N_2 = \tfrac{1}{4}(1+r)(1-s) &= \tfrac{1}{4}(1.5)(0.5) &= \tfrac{3}{16} &= 0.1875, \\ N_3 = \tfrac{1}{4}(1+r)(1+s) &= \tfrac{1}{4}(1.5)(1.5) &= \tfrac{9}{16} &= 0.5625, \\ N_4 = \tfrac{1}{4}(1-r)(1+s) &= \tfrac{1}{4}(0.5)(1.5) &= \tfrac{3}{16} &= 0.1875, \end{aligned} \tag{20}$$

and as a check $N_1 + N_2 + N_3 + N_4 = 0.0625 + 0.1875 + 0.5625 + 0.1875 = 1$, as (36) requires.

Notice what was *not* used. Nothing in (20) mentions D's corners, or its size, or where it sits. At this point the element could be *any* quadrilateral anywhere in the mesh and these four numbers would be identical. The physical geometry has not entered yet — and it enters only in Steps (ii) and (iii), through the corner coordinates.

Step (ii) — the x coordinate. The corner values are $x_1 = 0$, $x_2 = 3$, $x_3 = 4$, $x_4 = 1$:

$$\begin{aligned} x &= N_1 x_1 + N_2 x_2 + N_3 x_3 + N_4 x_4 \\ &= 0.0625\,(0) \;+\; 0.1875\,(3) \;+\; 0.5625\,(4) \;+\; 0.1875\,(1) \\ &= 0 \;+\; 0.5625 \;+\; 2.25 \;+\; 0.1875 \\ &= 3. \end{aligned} \tag{21}$$

Step (iii) — the y coordinate. The corner values are $y_1 = 0$, $y_2 = 0$, $y_3 = 4$, $y_4 = 5$:

$$\begin{aligned} y &= N_1 y_1 + N_2 y_2 + N_3 y_3 + N_4 y_4 \\ &= 0.0625\,(0) \;+\; 0.1875\,(0) \;+\; 0.5625\,(4) \;+\; 0.1875\,(5) \\ &= 0 \;+\; 0 \;+\; 2.25 \;+\; 0.9375 \\ &= 3.1875. \end{aligned} \tag{22}$$

So $(r, s) = (0.5, 0.5)$ sits at the physical point (3, 3.1875) — the entry in row D of Table 7. And N_3 itself is *unchanged* at 0.5625 — because N_3 is a function of (r, s) alone and never sees the element's shape. For A, B and C the reference square is a convenience; for D it is the only thing that works.

What the x, y coordinates are *for*

The physical description of an element carries exactly three things:

location	where the element sits in the mesh
size	how big it is
shape	its aspect ratio and distortion

Those three are the *only* things that tell one element from another, and they are the only reason the "N_3 in x, y" column of Table 7 changed from row to row. The reference square carries **none** of them: in (r, s) every element in the mesh is literally the same square, and $N_1, \ldots, N_4$ cannot tell them apart.

So where did location, size and shape go? Into the map — and from the map into exactly two places:

- **The corner coordinates** x_a, y_a, used by $x = \sum_a N_a x_a$ whenever an actual physical position is wanted. That is all of Steps (ii) and (iii), and nothing else needs them.

- **The Jacobian** J, which carries size and shape into every integral and every derivative:

$$\mathrm{d}\Omega = |J|\,\mathrm{d}r\,\mathrm{d}s, \qquad \frac{\partial N_a}{\partial x} = \frac{\partial N_a}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial N_a}{\partial s}\frac{\partial s}{\partial x}. \tag{23}$$

And J is where the difference between a tidy element and a distorted one actually shows up:

$$\text{rectangle } a \times b: \quad J = \frac{ab}{4} \ \text{(constant)}, \qquad \text{element } D: \quad J = \frac{5}{4} + \frac{r}{8} - \frac{s}{4} \ \text{(varies)}. \tag{24}$$

For the rectangle J is one number for the whole element — which is why (42) came out in closed form. For D it ranges from 3 to 4 across the four corners, so it must be re-evaluated at every quadrature point. (Integrating it recovers the area: $\int_{-1}^{1}\int_{-1}^{1} J\,\mathrm{d}r\,\mathrm{d}s = 14$, which is the area of D.)

The division of labour. (r, s) holds the *interpolation* — how the four nodal values are blended, identical for every element. x, y holds the *geometry* — location, size, shape, compressed into the corner coordinates and the Jacobian. The two meet only inside the integral, and nowhere else.

The map only runs one way — and that is all that is needed

Notice the asymmetry in what was just done. Given (r, s), the four contributions N_1, N_2, N_3, N_4 are known *immediately* — (20) is substitution, nothing more — and the physical point follows by (21)–(22). Going the other way is not like that at all. For element D the forward map is

$$x(r, s) = 2 + \tfrac{3}{2}r + \tfrac{1}{2}s, \qquad y(r, s) = \tfrac{9}{4} - \tfrac{1}{4}r + \tfrac{9}{4}s - \tfrac{1}{4}rs, \tag{25}$$

so recovering (r, s) from a given (x, y) means solving *two coupled nonlinear equations* — there is an rs term, so the system is not linear. Eliminating r leaves a **quadratic** in s, and a quadratic has two roots. For the point (3, 3.1875) they are

$$(r, s) = \left(\tfrac{1}{2}, \tfrac{1}{2}\right) \ \checkmark \text{ inside } [-1, 1]^2, \qquad (r, s) = \left(\tfrac{19}{2}, -\tfrac{53}{2}\right) \ \times \text{ outside — discard.} \tag{26}$$

So the inverse exists here, but it costs a quadratic solve *and* a test to throw away the spurious root. For a curved or three-dimensional element there is no closed form at all and one iterates with Newton's method.

Why this does not matter. The finite element method never asks the inverse question. Every quantity it needs is an *integral over the element*, and (41) turns each one into an integral over the reference square: you place Gauss points in (r, s), evaluate N_a and its derivatives there, and map *forward* to (x, y) only if the integrand needs it. The traffic is one-way by construction, which is exactly why only the easy direction was ever built.

The one place the inverse does come up is post-processing: "what is the solution at this particular physical point?" — for a sensor location, say. A code answers it by finding which element contains the point and then inverting the map numerically, which is why probing an arbitrary point is more expensive than reading a nodal value, where the answer is just ξ_j.

Read the number correctly. N_3 belongs to **node** 3 — the subscript names a *node*, never a point. Evaluating it somewhere returns the *weight that node 3's value carries at that location*:

$$u_h(P) = N_1(P)\,u_1 + N_2(P)\,u_2 + \underbrace{N_3(P)\,u_3}_{\text{node 3's share}} + N_4(P)\,u_4.$$

So the 0.5625 in Table 7 is not "the value of N_3 at $(0.5, 0.5)$" as though it were a property of that point — it is *how much node* 3 *contributes there*. On element B, node 3 sits physically at $(3, 2)$ and contributes 0.5625 to the point $(2.50, 1.50)$. Which node a shape function belongs to is fixed; changing between x, y and (r, s) only changes the coordinates used to *locate* the evaluation point.

In Table 7 the region, the x, y formula and both "sits at" columns change from element to element; the two share columns do not. As counted above, a mesh of 10,000 elements would need 40,000 formulas in x, y and needs exactly **four** in (r, s). Everything about where the element is and how big it is has moved out of the shape functions and into the map. Three consequences:

- **You write the formulas once.** (34) is hard-coded in the program. Nothing about it changes from element to element, or from mesh to mesh.
- **Every integral gets the same limits.** The element matrix (41) becomes $\int_{-1}^{1}\int_{-1}^{1}(\cdots)\,|J|\,\mathrm{d}r\,\mathrm{d}s$ for *every* element, so one fixed Gauss rule — the same points, the same weights — integrates the whole mesh. Working in x, y you would need a different set of quadrature points for every element.
- **It is the only practical route for element D.** For four arbitrary corners the bilinear function in x, y has no tidy closed form, but the map still works: use the *same* N_a to carry the geometry, $x(r,s) = \sum_a N_a(r,s)\,x_a$ and $y(r,s) = \sum_a N_a(r,s)\,y_a$. That is what **isoparametric** means — same shapes for the geometry as for the unknown — and it is why real meshes may use distorted quads at all.

Put the other way round: the reference square is a change of variables, nothing deeper. It does not alter φ_j or the method — (7) is still what is being computed. It only spares you from re-deriving the same four polynomials once per element.

Step 4 — this is where φ becomes N

Definition. N_a is the global basis function φ_j, restricted to one element, and rewritten in the reference coordinates:

$$N_a(r,s) \;\coloneqq\; \varphi_j\big|_K\big(x(r,s),\, y(r,s)\big), \qquad a = \text{local number of global node } j \text{ in } K.$$

That is the whole of it. N is not a new object — it is φ seen on one element, in borrowed coordinates.

And the nodal condition comes along with it. Restrict (4) to K's own four corners: φ_j is 1 at its own corner and 0 at the other three, so in reference coordinates

$$\boxed{N_a(\text{node } b) = \delta_{ab}, \qquad a, b = 1, 2, 3, 4} \tag{27}$$

which is (4) with the sum over M global nodes narrowed to the 4 local ones.

Step 5 — now solve for the four N's

By Step 2 each N_a is bilinear, so write

$$N(r,s) = c_0 + c_1 r + c_2 s + c_3\, rs, \tag{28}$$

and evaluate at the four corners $(-1,-1)$, $(1,-1)$, $(1,1)$, $(-1,1)$:

$$\begin{aligned} N(-1,-1) &= c_0 - c_1 - c_2 + c_3, & N(1,-1) &= c_0 + c_1 - c_2 - c_3, \\ N(1,1) &= c_0 + c_1 + c_2 + c_3, & N(-1,1) &= c_0 - c_1 + c_2 - c_3. \end{aligned} \tag{29}$$

Setting these equal to the values demanded by (27) gives four linear systems, one per node.

Node 1: $N_1 = 1$ at node 1, 0 at nodes 2, 3, 4.

$$\begin{cases} c_0 - c_1 - c_2 + c_3 = 1 \\ c_0 + c_1 - c_2 - c_3 = 0 \\ c_0 + c_1 + c_2 + c_3 = 0 \\ c_0 - c_1 + c_2 - c_3 = 0 \end{cases} \quad\Longrightarrow\quad c_0 = \tfrac{1}{4},\ c_1 = -\tfrac{1}{4},\ c_2 = -\tfrac{1}{4},\ c_3 = \tfrac{1}{4}, \tag{30}$$

$$N_1 = \tfrac{1}{4}\big(1 - r - s + rs\big) = \tfrac{1}{4}(1-r)(1-s).$$

Node 2: $N_2 = 1$ at node 2, 0 at nodes 1, 3, 4.

$$\begin{cases} c_0 - c_1 - c_2 + c_3 = 0 \\ c_0 + c_1 - c_2 - c_3 = 1 \\ c_0 + c_1 + c_2 + c_3 = 0 \\ c_0 - c_1 + c_2 - c_3 = 0 \end{cases} \quad\Longrightarrow\quad c_0 = \tfrac{1}{4},\ c_1 = \tfrac{1}{4},\ c_2 = -\tfrac{1}{4},\ c_3 = -\tfrac{1}{4}, \tag{31}$$

$$N_2 = \tfrac{1}{4}\big(1 + r - s - rs\big) = \tfrac{1}{4}(1+r)(1-s).$$

Node 3: $N_3 = 1$ at node 3, 0 at nodes 1, 2, 4.

$$\begin{cases} c_0 - c_1 - c_2 + c_3 = 0 \\ c_0 + c_1 - c_2 - c_3 = 0 \\ c_0 + c_1 + c_2 + c_3 = 1 \\ c_0 - c_1 + c_2 - c_3 = 0 \end{cases} \quad\Longrightarrow\quad c_0 = \tfrac{1}{4},\ c_1 = \tfrac{1}{4},\ c_2 = \tfrac{1}{4},\ c_3 = \tfrac{1}{4}, \tag{32}$$

$$N_3 = \tfrac{1}{4}\big(1 + r + s + rs\big) = \tfrac{1}{4}(1+r)(1+s).$$

Node 4: $N_4 = 1$ at node 4, 0 at nodes 1, 2, 3.

$$\begin{cases} c_0 - c_1 - c_2 + c_3 = 0 \\ c_0 + c_1 - c_2 - c_3 = 0 \\ c_0 + c_1 + c_2 + c_3 = 0 \\ c_0 - c_1 + c_2 - c_3 = 1 \end{cases} \quad\Longrightarrow\quad c_0 = \tfrac{1}{4},\ c_1 = -\tfrac{1}{4},\ c_2 = \tfrac{1}{4},\ c_3 = -\tfrac{1}{4}, \tag{33}$$

$$N_4 = \tfrac{1}{4}\big(1 - r + s - rs\big) = \tfrac{1}{4}(1-r)(1+s).$$

Collecting the four:

$$\boxed{\begin{aligned} N_1(r,s) &= \tfrac{1}{4}(1-r)(1-s), & N_2(r,s) &= \tfrac{1}{4}(1+r)(1-s), \\ N_3(r,s) &= \tfrac{1}{4}(1+r)(1+s), & N_4(r,s) &= \tfrac{1}{4}(1-r)(1+s). \end{aligned}} \tag{34}$$

There is a shortcut worth seeing, because it explains the product form: each N_a must vanish on the *two edges* not touching node a. Those edges are $r = \mp 1$ and $s = \mp 1$, so N_a must contain both factors; normalise by dividing by its value at its own node, which is 4. That is (34) without solving anything. A compact way to write all four:

$$N_a(r,s) = \tfrac{1}{4}\big(1 + r_a r\big)\big(1 + s_a s\big), \qquad (r_a, s_a) = \text{coordinates of node } a. \tag{35}$$

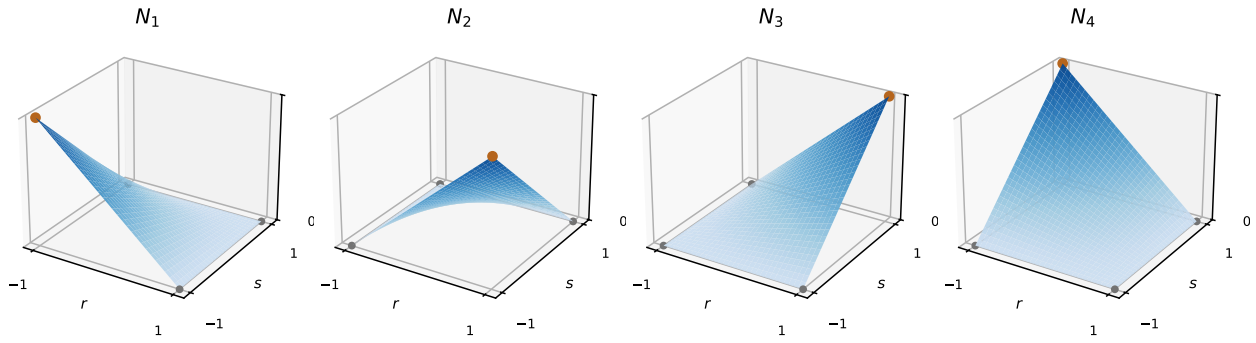


Figure 9: The four bilinear shape functions on the reference square. Each is 1 at its own node (orange) and 0 at the other three (grey) — which is exactly (27), the lecture's nodal table written for a quadrilateral.

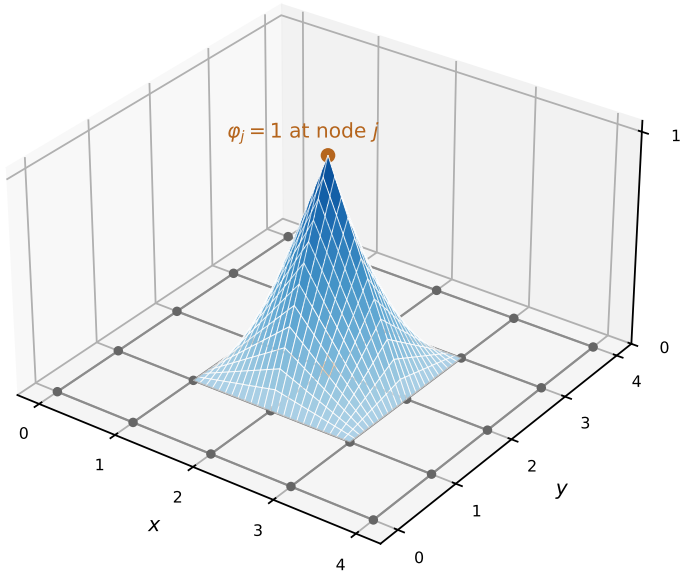


Figure 10: The same four shapes, assembled. On the 16-element mesh of §3 Step 1 ($5 \times 5 = 25$ nodes), the global basis function φ_j is a **hat** standing on exactly four elements and flat — identically zero — on the other twelve. Its four faces are N_3, N_4, N_2, N_1 of those four elements, which is (38) drawn. This is the object the lecturer means by φ_j; the four panels of Figure 9 are the pieces it is cut from.

Two identities that follow immediately (both verified symbolically):

$$\sum_{a=1}^{4} N_a(r,s) = 1 \qquad \text{(partition of unity)}, \tag{36}$$

$$\sum_{a=1}^{4} N_a(r,s)\, r_a = r, \qquad \sum_{a=1}^{4} N_a(r,s)\, s_a = s \qquad \text{(linear fields reproduced exactly)}. \tag{37}$$

(36) is the rigid-body (constant) mode: a uniform nodal value gives a uniform field. (37) is the constant-gradient mode. Together they are the patch test, and they are why the element converges at all — without them $\|u - \Pi_h u\| \not\to 0$ and Céa's lemma delivers nothing.

4. φ versus N: the picture

N_a lives on **one element**, in local coordinates. The lecture's φ_j lives on **all of** Ω, in global coordinates, one per global node. The relation:

φ_j restricted to element e $=$ N_a, where a is the *local* number of global node j inside e.

In a structured quad mesh an interior node belongs to 4 elements, so φ_j is 4 local N's glued together, and zero everywhere else.

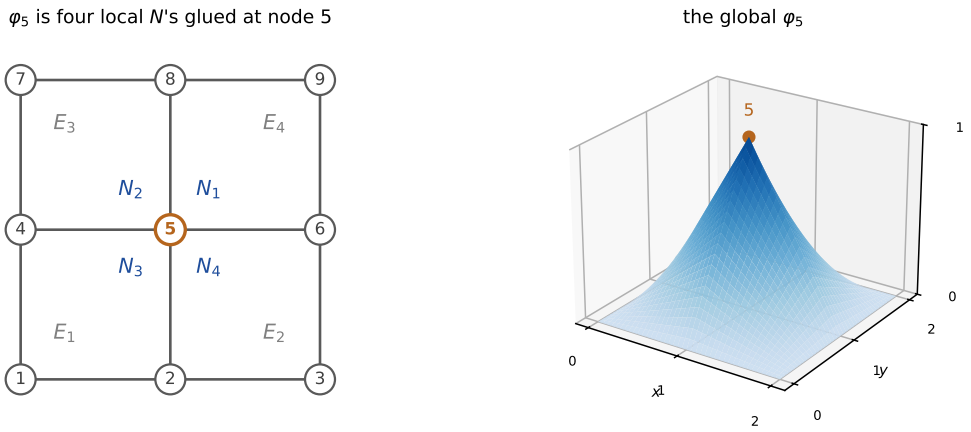


Figure 11: Left: node j is shared by four elements, and in each one it carries a different *local* number — so φ_j is built from N_3, N_4, N_2, N_1 of those four elements respectively. Right: the resulting φ_j, the function the lecturer actually means.

Writing that φ out explicitly

Take the patch in Figure 11: four unit squares on $[0,2]^2$, node 5 at the centre $(1,1)$. Each element maps to the reference square by $r = 2(x - x_e) - 1$, $s = 2(y - y_e) - 1$, where (x_e, y_e) is its lower-left corner. Substituting that into the N named in each quadrant:

element	region	node 5 is local	φ_5 there	
E_1	$[0,1]\times[0,1]$	3	$N_3=\frac{1}{4}(1+r)(1+s)$	$\xrightarrow{r=2x-1,\ s=2y-1}$ $\boldsymbol{xy}$
E_2	$[1,2]\times[0,1]$	4	$N_4=\frac{1}{4}(1-r)(1+s)$	$\xrightarrow{r=2x-3,\ s=2y-1}$ $\boldsymbol{(2-x)\,y}$
E_3	$[0,1]\times[1,2]$	2	$N_2=\frac{1}{4}(1+r)(1-s)$	$\xrightarrow{r=2x-1,\ s=2y-3}$ $\boldsymbol{x\,(2-y)}$
E_4	$[1,2]\times[1,2]$	1	$N_1=\frac{1}{4}(1-r)(1-s)$	$\xrightarrow{r=2x-3,\ s=2y-3}$ $\boldsymbol{(2-x)(2-y)}$

Table 10: φ_5 written out on each of the four elements of Figure 11.

so the global basis function of node 5 is, in one piece,

$$\varphi_5(x,y)=\begin{cases} x\,y, & 0\le x\le 1,\ 0\le y\le 1 \quad (E_1),\\ (2-x)\,y, & 1\le x\le 2,\ 0\le y\le 1 \quad (E_2),\\ x\,(2-y), & 0\le x\le 1,\ 1\le y\le 2 \quad (E_3),\\ (2-x)(2-y), & 1\le x\le 2,\ 1\le y\le 2 \quad (E_4),\\ 0, & \text{everywhere else.}\end{cases} \tag{38}$$

Equivalently, in closed form, $\varphi_5(x,y)=\hat{h}(x-1)\,\hat{h}(y-1)$ with $\hat{h}(t)=\max(0,1-|t|)$ — a product of two 1D hats. (Checked: the four pieces agree with this at interior sample points.)

Two things to read off (38).

- **It is continuous, so $\varphi_5\in H^1$.** Across the interior edge $x=1$: from E_1 it is $xy\to y$, from E_2 it is $(2-x)y\to y$. Equal. Likewise across $y=1$. This is the C^0 conformity of §2, now visible as algebra.
- **Its gradient jumps, and that is allowed.** At $x=1$, $\partial_x\varphi_5=y$ from the left and $-y$ from the right — a jump of $2y$. The lecture only ever required $\iint\|\nabla v\|^2<\infty$, and a finite jump is square integrable. A jump in φ itself would not be.
- Each piece is a different formula, which is the practical reason nobody assembles φ — whereas the four N_a in (34) are the same for every element in the mesh.

In practice you never build the right-hand picture of Figure 11, nor (38). You loop over elements and use the left-hand one. The φ_j exists only implicitly, inside the connectivity table — which is the honest answer to "where is φ in my code?" It is the `IEN`/`LM` array.

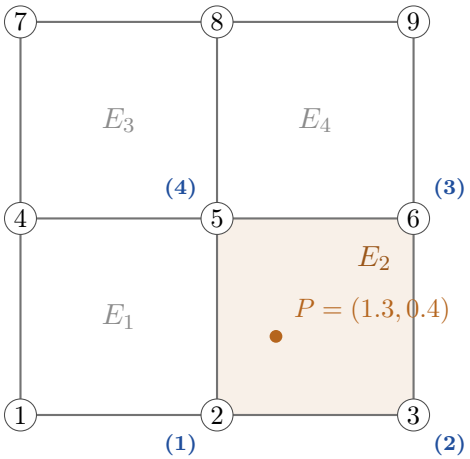
5. Worked example: a mesh with four elements

Everything above is one element. Here is what φ and N actually *equal*, as numbers, when there is more than one.

The mesh. Four unit squares on $[0,2]\times[0,2]$: 9 global nodes, 4 elements. Global node numbers are circled, element numbers are in the centres, and the orange dot is the point

$$P=(1.3,\ 0.4)$$

at which we will evaluate everything.



(a) The φ values — nine numbers, five of them zero

Each global φ_j is the product of two 1D hats, $\varphi_j(x,y) = \hat{h}(x-x_j)\,\hat{h}(y-y_j)$ with $\hat{h}(t) = \max(0, 1-|t|)$. Evaluating all nine at $P = (1.3, 0.4)$:

global node j	position	$\varphi_j(P)$	
1	$(0,0)$	0	P is more than one element away
2	$(1,0)$	**0.42**	
3	$(2,0)$	**0.18**	
4	$(0,1)$	0	
5	$(1,1)$	**0.28**	
6	$(2,1)$	**0.12**	
7	$(0,2)$	0	
8	$(1,2)$	0	
9	$(2,2)$	0	
		$\sum = 1.00$	partition of unity, (36)

Table 11: All nine φ_j evaluated at $P = (1.3, 0.4)$. Only the four corners of the element containing P are non-zero.

Read this off the sketch. The only non-zero φ's are those of nodes $2, 3, 6, 5$ — exactly **the four corners of the shaded element** E_2, the one containing P. Every other node is too far away: its hat has already fallen to zero before reaching P. That is "small support", in numbers.

(b) The N values — four numbers, in element E_2 only

E_2 spans $x \in [1,2]$, $y \in [0,1]$, so the map to the reference square is $r = 2x - 3$, $s = 2y - 1$, giving at P

$$r = 2(1.3) - 3 = -0.40, \qquad s = 2(0.4) - 1 = -0.20.$$

Substituting into (34):

$$N_1 = \tfrac{1}{4}(1-r)(1-s) = \tfrac{1}{4}(1.4)(1.2) = \mathbf{0.42}, \qquad N_2 = \tfrac{1}{4}(1+r)(1-s) = \tfrac{1}{4}(0.6)(1.2) = \mathbf{0.18},$$
$$N_3 = \tfrac{1}{4}(1+r)(1+s) = \tfrac{1}{4}(0.6)(0.8) = \mathbf{0.12}, \qquad N_4 = \tfrac{1}{4}(1-r)(1+s) = \tfrac{1}{4}(1.4)(0.8) = \mathbf{0.28},$$

and again $\sum_a N_a = 1.00$.

(c) They are the same four numbers

The connectivity of E_2 — the blue **(a)** labels in the sketch — is local $1, 2, 3, 4 \;\mapsto$ global $2, 3, 6, 5$. Lining the two tables up:

local a	global j	$N_a(r, s)$	$\varphi_j(P)$	
1	2	0.42	0.42	✓
2	3	0.18	0.18	✓
3	6	0.12	0.12	✓
4	5	0.28	0.28	✓

Table 12: The four N values and the four non-zero φ values are the same numbers, matched by the connectivity.

So, to answer the question directly. With several elements:

- There are $M = 9$ functions φ_j, one per global node, each defined over the *whole* mesh. At any given point, **at most four are non-zero** — the corners of the element you are standing in.
- There are 4 functions N_a, defined only on the reference square, and **the same four formulas are reused for every element** in the mesh.
- At a point inside element e, the four non-zero φ values *are* the four N values, matched up by the connectivity table.

φ is one function per unknown spread over the mesh; N is one formula per corner reused everywhere. Same numbers.

This is also why $\boldsymbol{K}$ is sparse. A_{ij} is non-zero only when φ_i and φ_j overlap, i.e. when nodes i and j share an element. In the sketch, node 1 and node 9 never share one, so $A_{1,9} = 0$ — without computing anything.

6. Why each side uses what it uses

The two views are not a matter of taste. Each is the only one that works for what that side needs to do.

Why the mathematician insists on φ

- The weak form $\iint_\Omega \nabla u \cdot \nabla v \, \mathrm{d}\Omega$ is **one integral over** Ω, and V_h is **one space**.
- Every theorem in the course is a statement about *functions on* Ω, not about elements:

statement	what it quantifies over
$u_h \in V_h$	a single function on all of Ω
Galerkin orthogonality $B(u - u_h, v) = 0 \; \forall v \in V_h$	every v in the whole space
Céa $\|u - u_h\| \leq \frac{\beta^*}{\beta_*} \|u - v\|$	norms over all of Ω
Lax–Milgram	existence/uniqueness in V

- **You cannot even write "$u_h \in V_h$" in local coordinates.** $N_a(r,s)$ is defined on $[-1,1]^2$; it is not a function on Ω at all until you say *which element* and compose with the map. Conformity, $V_h \subset V$, is a statement about continuity *between* elements — precisely the thing local coordinates cannot see.

Why the engineer never builds φ

- You loop over elements, compute $\boldsymbol{k}^e$ from the N's, and scatter into $\boldsymbol{K}$. At no point is a global φ_j ever formed.
- It would be wasteful: φ_j is zero on all but 4 of the thousands of elements, so assembling it explicitly means storing and integrating mostly zeros.
- It would also be awkward: φ_j has a different algebraic expression on each of its four elements, whereas $N_a(r,s)$ is the *same* four formulas for every element in the mesh. All the geometry is pushed into the Jacobian.
- **So where is φ in the code?** In the connectivity table — the `IEN`/`LM` array. The statement "global node j is local node a of element e" *is* the statement $\varphi_j|_{K_e} = N_a$. The array is the only place φ is recorded, and it is recorded as an integer.

The identity that makes both legal

The two views meet in one equation. Start from the lecture's definition,

$$A_{ij} = B(\varphi_i, \varphi_j) = \iint_\Omega \nabla\varphi_i \cdot \nabla\varphi_j \,\mathrm{d}\Omega, \tag{39}$$

and split the integral element by element — legal because $\Omega = \bigcup_e K_e$ and the elements overlap only on edges, which have zero area:

$$\boxed{A_{ij} = \sum_e \iint_{K_e} \nabla\varphi_i \cdot \nabla\varphi_j \,\mathrm{d}\Omega = \sum_e \underbrace{\iint_{K_e} \nabla N_a \cdot \nabla N_b \,\mathrm{d}\Omega}_{k^e_{ab}}} \tag{40}$$

where a, b are the local numbers of global nodes i, j in element e, and the term is simply absent if either node does not belong to e.

Read (40) from both ends. The left-hand side is the mathematician's *definition* of the stiffness matrix — a single integral of global basis functions, the object Céa and Lax–Milgram talk about. The right-hand side is the *loop you already write*. They are the same number.

Three consequences worth stating plainly:

- **Assembly is not an approximation.** It is an exact identity — just additivity of the integral. Nothing is lost going from φ to N.
- **The sum is short.** It runs only over elements containing *both* nodes i and j — at most two for a Q4 edge pair, four for $i = j$. That is why $\boldsymbol{K}$ is sparse, and it is a direct consequence of φ having small support.
- **Every theorem transfers for free.** Because the two sides are equal, anything proved about A_{ij} is automatically true of your assembled $\boldsymbol{K}$ — symmetry, positive definiteness, the Céa bound. You get the theory without ever leaving local coordinates.

This is the real reason the course is taught in φ and your code is written in N: the global view is the only one in which the theorems can be *stated*, and the local view is the only one in which the integrals can be *computed*. (40) is the bridge, and it is the single most important line in this note.

7. The element stiffness matrix

Evaluating k^e on an $a \times b$ rectangle

Map the reference square to the element: $x = \frac{a}{2}r$, $y = \frac{b}{2}s$, so $\partial_x = \frac{2}{a}\partial_r$, $\partial_y = \frac{2}{b}\partial_s$, and $\mathrm{d}\Omega = \frac{ab}{4}\,\mathrm{d}r\,\mathrm{d}s$. Then

$$k^e_{ab} = \int_{-1}^{1}\int_{-1}^{1}\left[\tfrac{4}{a^2}\partial_r N_a\,\partial_r N_b + \tfrac{4}{b^2}\partial_s N_a\,\partial_s N_b\right]\tfrac{ab}{4}\,\mathrm{d}r\,\mathrm{d}s, \tag{41}$$

which integrates (exactly, by hand or by 2×2 Gauss) to

$$\boldsymbol{k}^e = \frac{b}{6a}\begin{bmatrix} 2 & -2 & -1 & 1\\ -2 & 2 & 1 & -1\\ -1 & 1 & 2 & -2\\ 1 & -1 & -2 & 2\end{bmatrix} + \frac{a}{6b}\begin{bmatrix} 2 & 1 & -1 & -2\\ 1 & 2 & -2 & -1\\ -1 & -2 & 2 & 1\\ -2 & -1 & 1 & 2\end{bmatrix}, \tag{42}$$

and for a **square** ($b = a$) the two collapse to the familiar

$$\boxed{\boldsymbol{k}^e_{\text{square}} = \frac{1}{6}\begin{bmatrix} 4 & -1 & -2 & -1\\ -1 & 4 & -1 & -2\\ -2 & -1 & 4 & -1\\ -1 & -2 & -1 & 4\end{bmatrix}} \tag{43}$$

Sanity checks (all verified):

- **Symmetric.** Forced by (S): $B(u,v) = B(v,u)$.
- **Every row sums to zero.** That is (36): a constant nodal field has zero gradient, so $\boldsymbol{k}^e\mathbf{1} = \mathbf{0}$.
- **Eigenvalues** $0,\ \frac{2}{3},\ \frac{2}{3},\ 1$ — positive semi-definite with exactly *one* zero mode, the constant. The single zero is the free-floating element's rigid-body mode; it disappears once a Dirichlet node is constrained, which is precisely the lecture's V-ellipticity (VE) being restored by the boundary condition, and hence $\boldsymbol{A}$ becoming SPD.

8. The load vector, and the rest of the dictionary

$$b_j = L(\varphi_j) = \iint_\Omega f\varphi_j\,\mathrm{d}\Omega \qquad\Longrightarrow\qquad f^e_a = \iint_{K_e} f\,N_a\,\mathrm{d}\Omega, \tag{44}$$

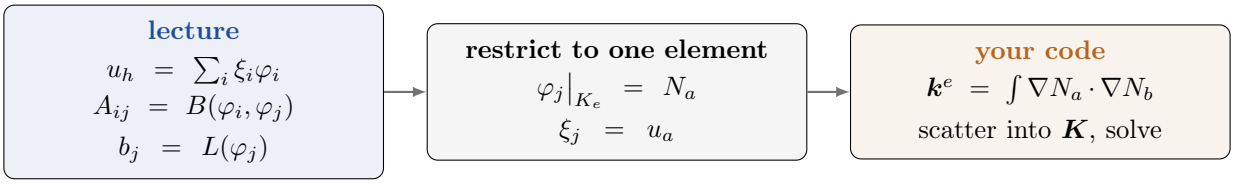
assembled the same way. For constant f on an $a\times b$ rectangle, $f^e_a = \frac{ab}{4}f$ at each of the four nodes — the load split evenly, which is the "consistent nodal load" your code reports.

And the theory transfers without being re-derived:

lecture result	what it tells you about your Q4 mesh
Lax–Milgram	the system has exactly one solution, and $\|u_h\| \le \widetilde{L}/\beta_*$
$\boldsymbol{A}$ is SPD under (S)+(VE)	you may use Cholesky / CG; no pivoting needed
Céa, $\|u-u_h\| \le \frac{\beta^*}{\beta_*}\|u-v\|$	your answer is within a fixed factor of the best Q4 fit
$\|u-\Pi_h u\| = O(h^p)$	bilinear gives $O(h)$ in H^1, $O(h^2)$ in L^2
regular mesh, $h_j/\rho_j \le \alpha$	don't let the quads become slivers, or the constant blows up

Table 13: Each lecture result, read as a statement about a Q4 mesh.

9. The whole bridge on one line



The one-sentence answer. φ and N are the same functions: φ is the global view (one per unknown, used to *state* the method and prove things about it), N is the local view (four per element, used to *compute*). ξ_i and your nodal unknown u_i are the same numbers. The bilinear N's are not an engineering invention bolted on afterwards — they are what the lecture's three requirements force on a quadrilateral.

Everything computed, not quoted: the N_a in (34) were solved from the nodal conditions (27), the identities (36)–(37) verified symbolically, and (42)–(43) integrated exactly, with symmetry, zero row sums and the eigenvalues checked. Script: `q4_figs.py` (figures), and the symbolic work is reproduced in the session log.